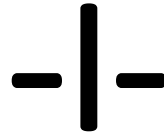


ANATOMY OF THE ARM



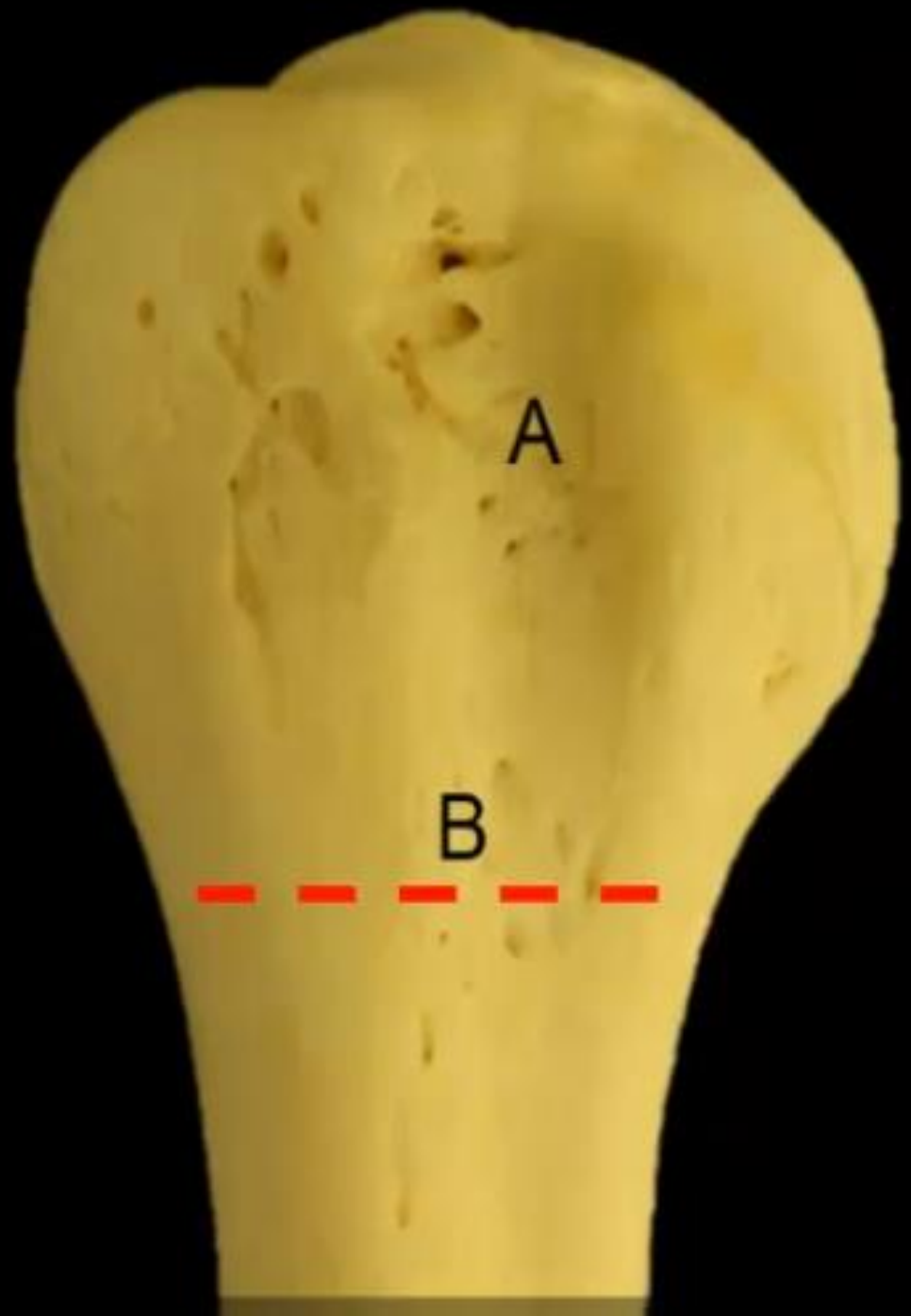
DR. AHMED AL-BAGHDADI, *Ph.D*

THE ARM

- Bone
- Fascia
- Anterior compartment
 - Biceps brachii
 - Brachialis
 - Coracobrachialis
- Posterior compartment
- Triceps
- ARTERIES & SUPERFICIAL VEINS
 - Axillary artery
 - Anastomosis around the scapula
 - Brachial artery
 - Anastomosis around the elbow

BONE





Lateral epicondyle

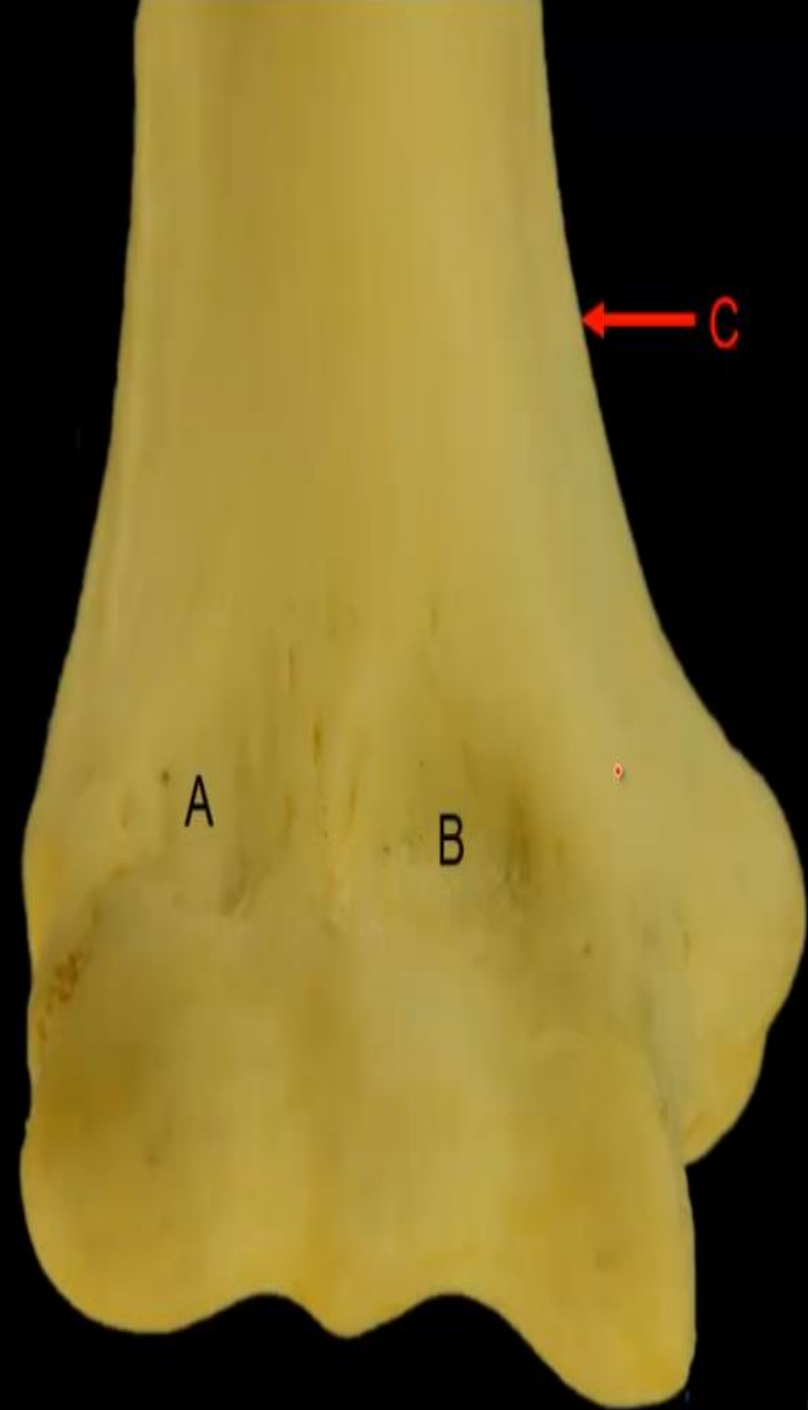


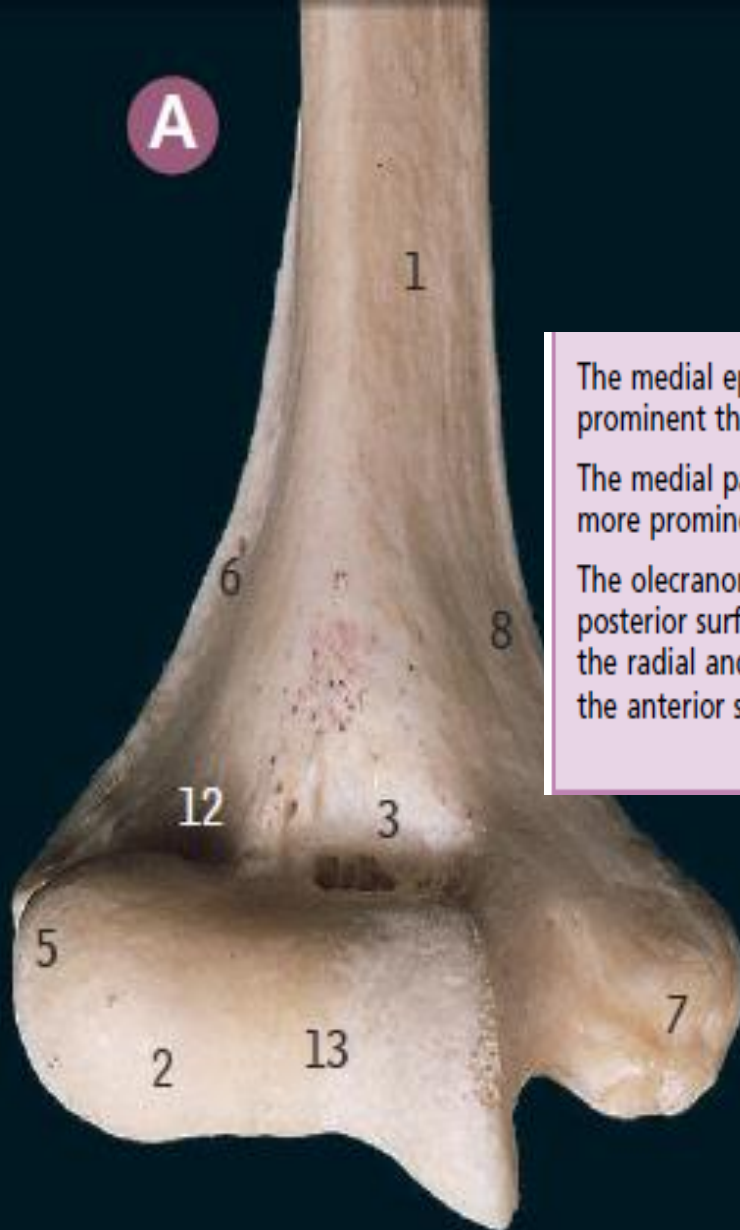
Medial epicondyle



Golfers elbow

©MMG 2000



A**B**

The medial epicondyle (7) is more prominent than the lateral (5).

The medial part of the trochlea (13) is more prominent than the lateral part.

The olecranon fossa (10) on the posterior surface is deeper than the radial and coronoid fossae on the anterior surface (12 and 3).



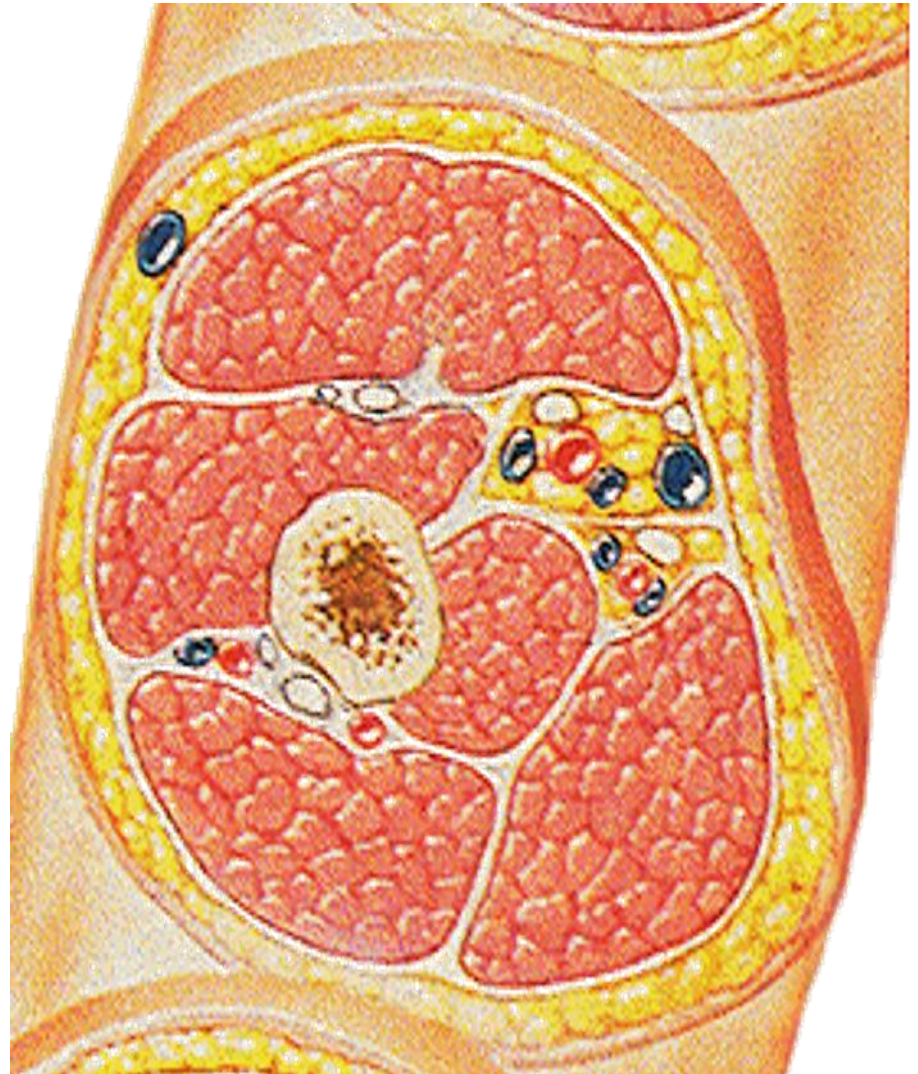


The trochlear notch (11) faces forwards, with the radial notch (9) on the lateral side.

FASCIA

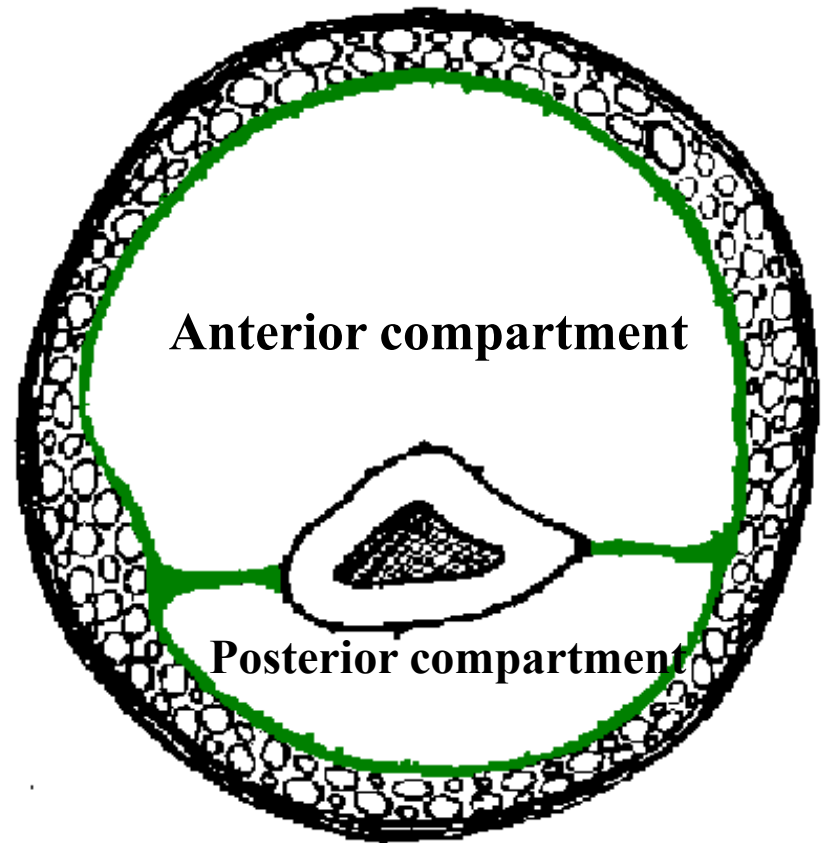
Deep Fascia

- The arm is enclosed in a sheath of deep fascia. from which
- two fascial intermuscular septa extend to the medial and lateral supracondylar ridges of the humerus



Compartments

- The medial and lateral intermuscular septa divide the arm into
- anterior and posterior fascial compartments.



COMPARTMENTS

ANTERIOR COMPARTMENT

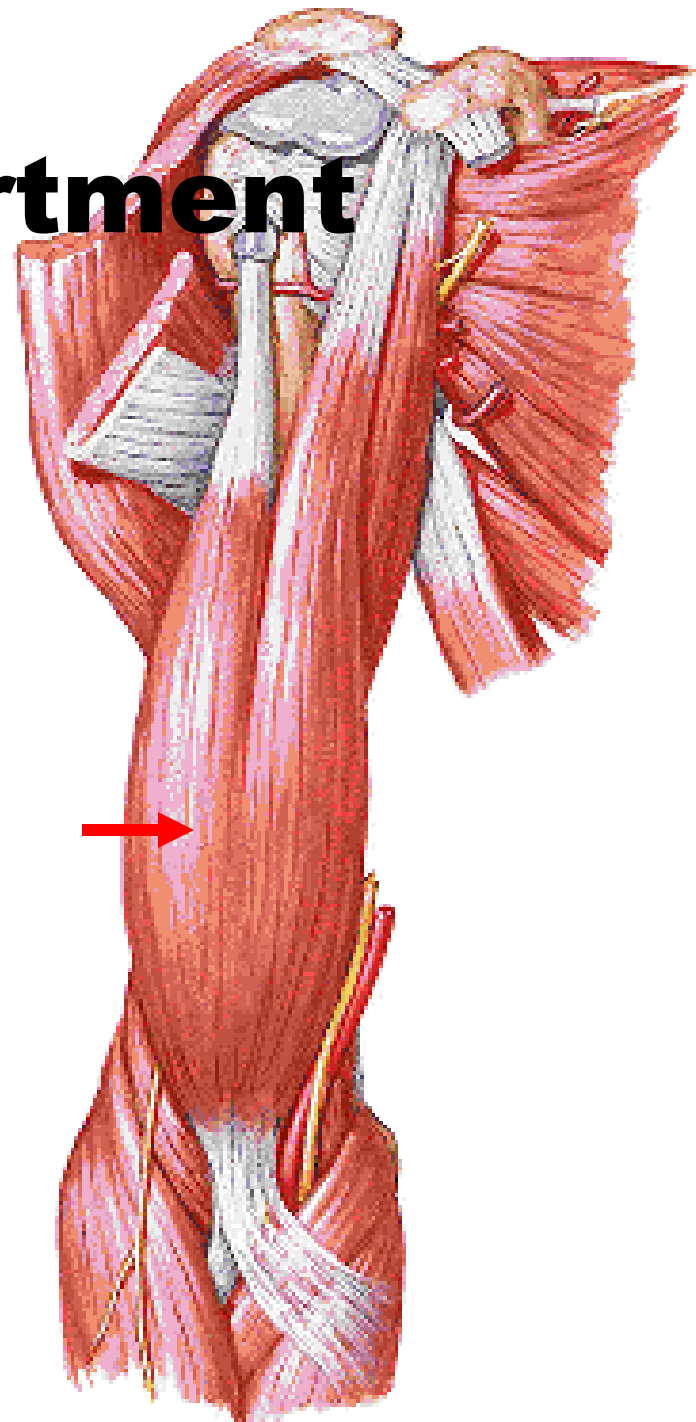
A. Muscles of front of arm

Muscle	Origin	Insertion	Nerve supply	Action.
1-Biceps Brachii	a. Long head: supraglenoid tubercle (intra-capsular extra-synovial). b. Short head: coracoid process.	Posterior part of radial tuberosity & deep fascia of upper part of medial side of forearm.	All muscles of front of arm are supplied by musculocutaneous nerve except lateral ½ of	Flexion of supinated forearm. Supination of flexed pronated forearm at radioulnar joints. • Weak flexor of shoulder.
2- Coracobrachialis	Coracoid process (with short head of biceps)	Middle of medial surface of humerus	brachialis (supplied by radial nerve i.e. double	Flexion & adduction of shoulder.
3- Brachialis	Lower 1/ 2 of humerus	Ulnar tuberosity.	nerve supply)	Main flexor of elbow.

Anterior compartment

contains **THREE** muscles

□ Biceps brachii

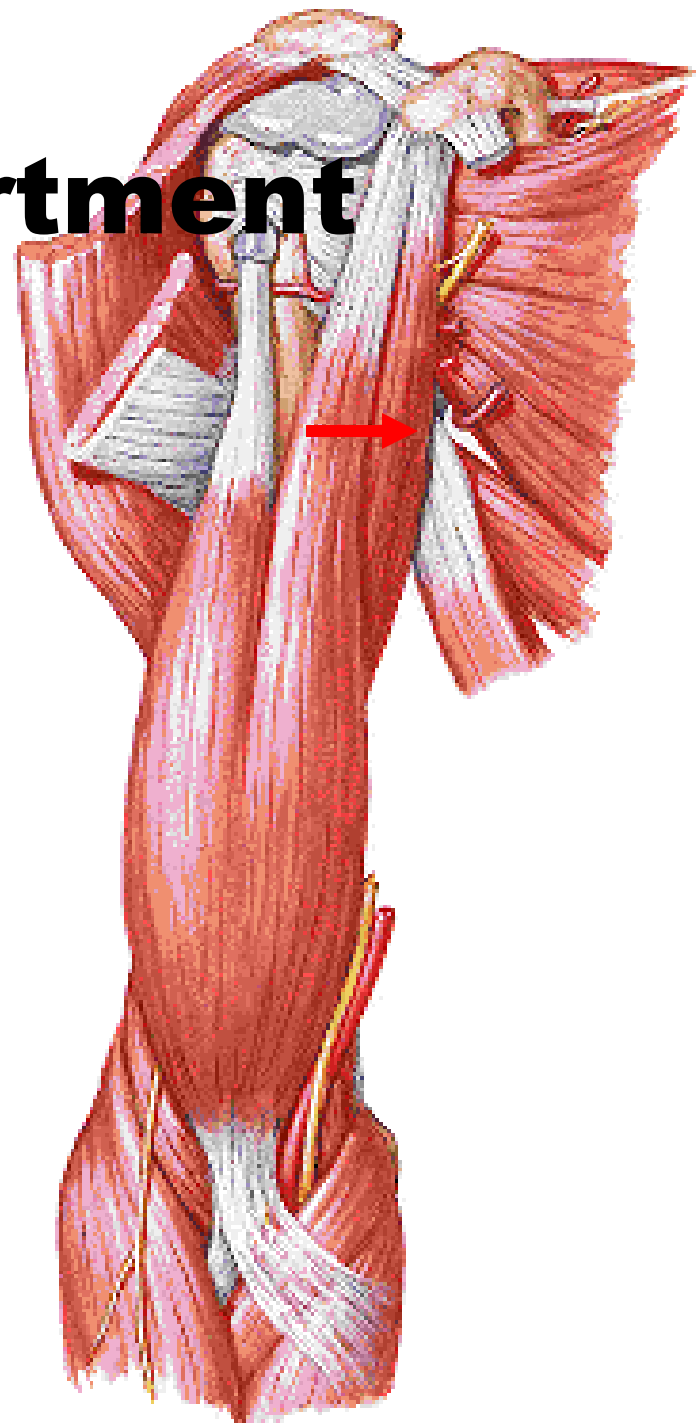


Anterior compartment

contains **THREE** muscles

□ Biceps brachii

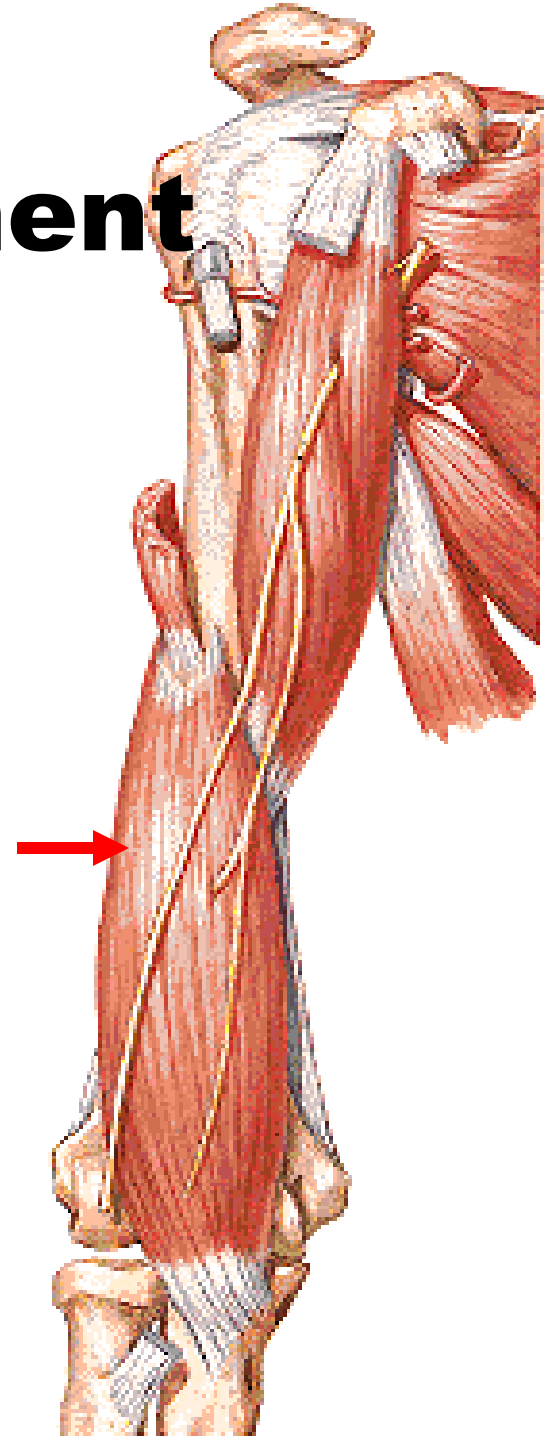
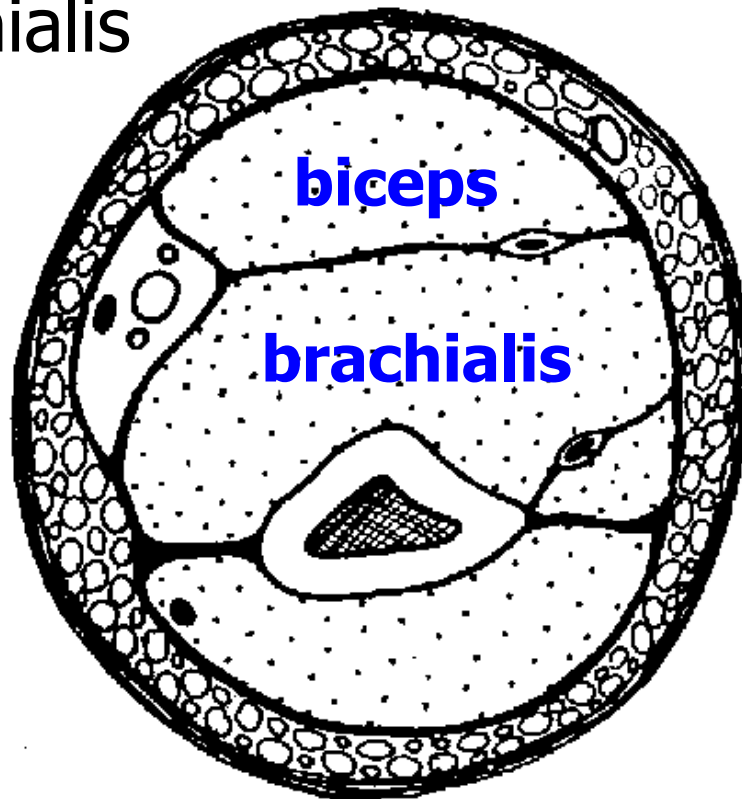
□ Coracobrachialis



Anterior compartment

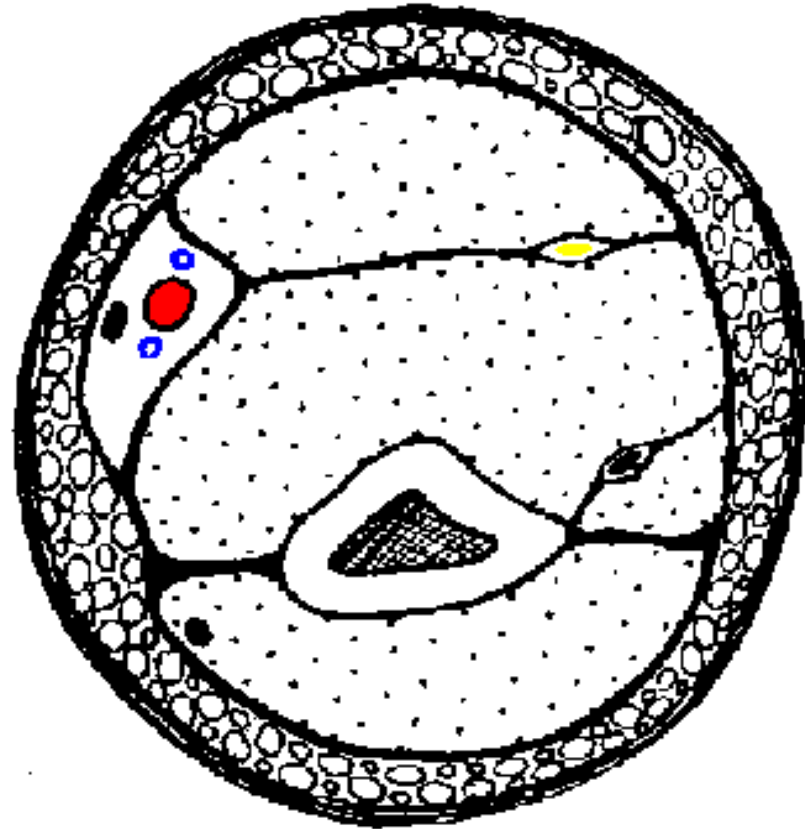
contains **THREE** muscles

- Biceps brachii
- Coracobrachialis
- Brachialis



Anterior compartment

- nerve of the compartment is
musculocutaneous nerve
- artery of compartment is
brachial artery



Biceps brachii

□ Origin

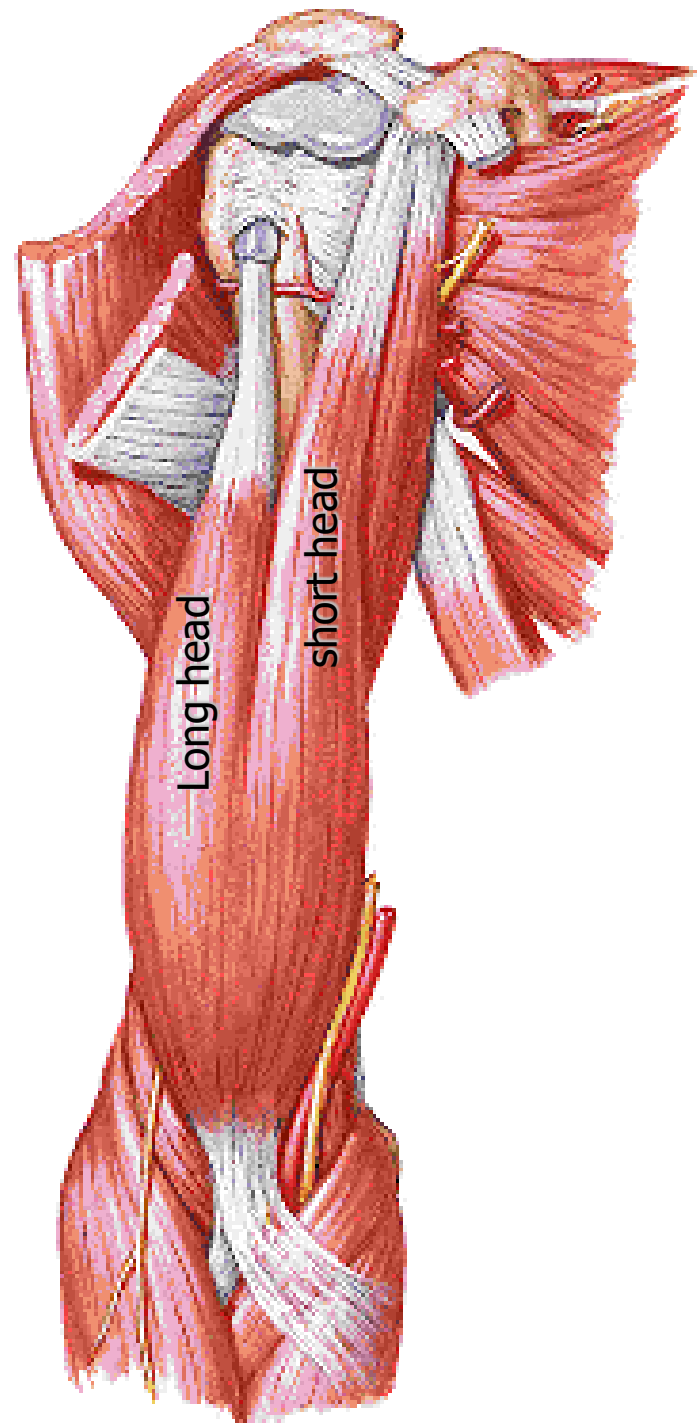
has 2 heads (as its name implies)

- long head

(which is lateral, L. for L.)

- short head

which is medial



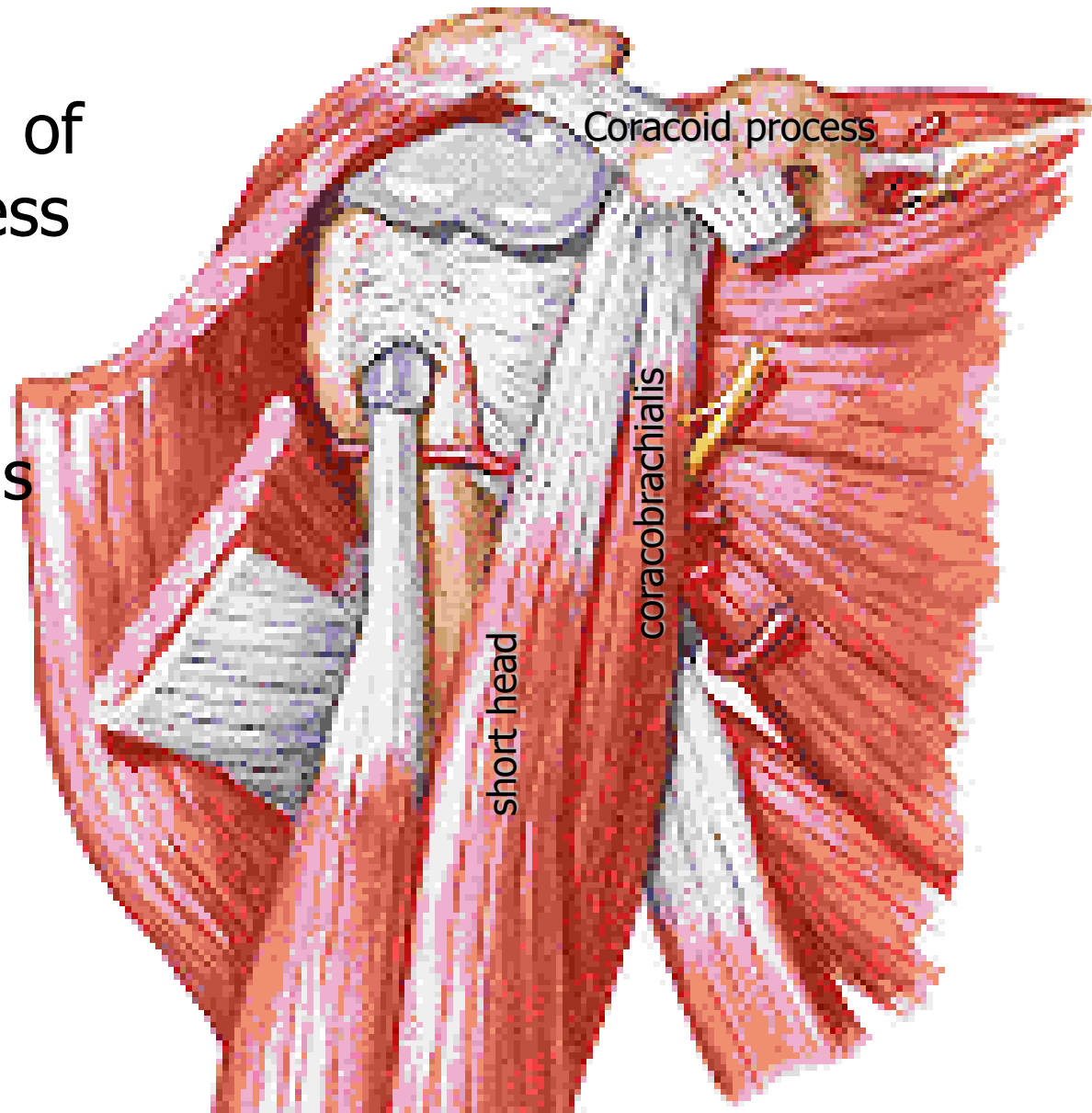
Long head of biceps brachii



- emerges at the top of the intertubercular sulcus
- surrounded by a tube-like extension of synovial membrane

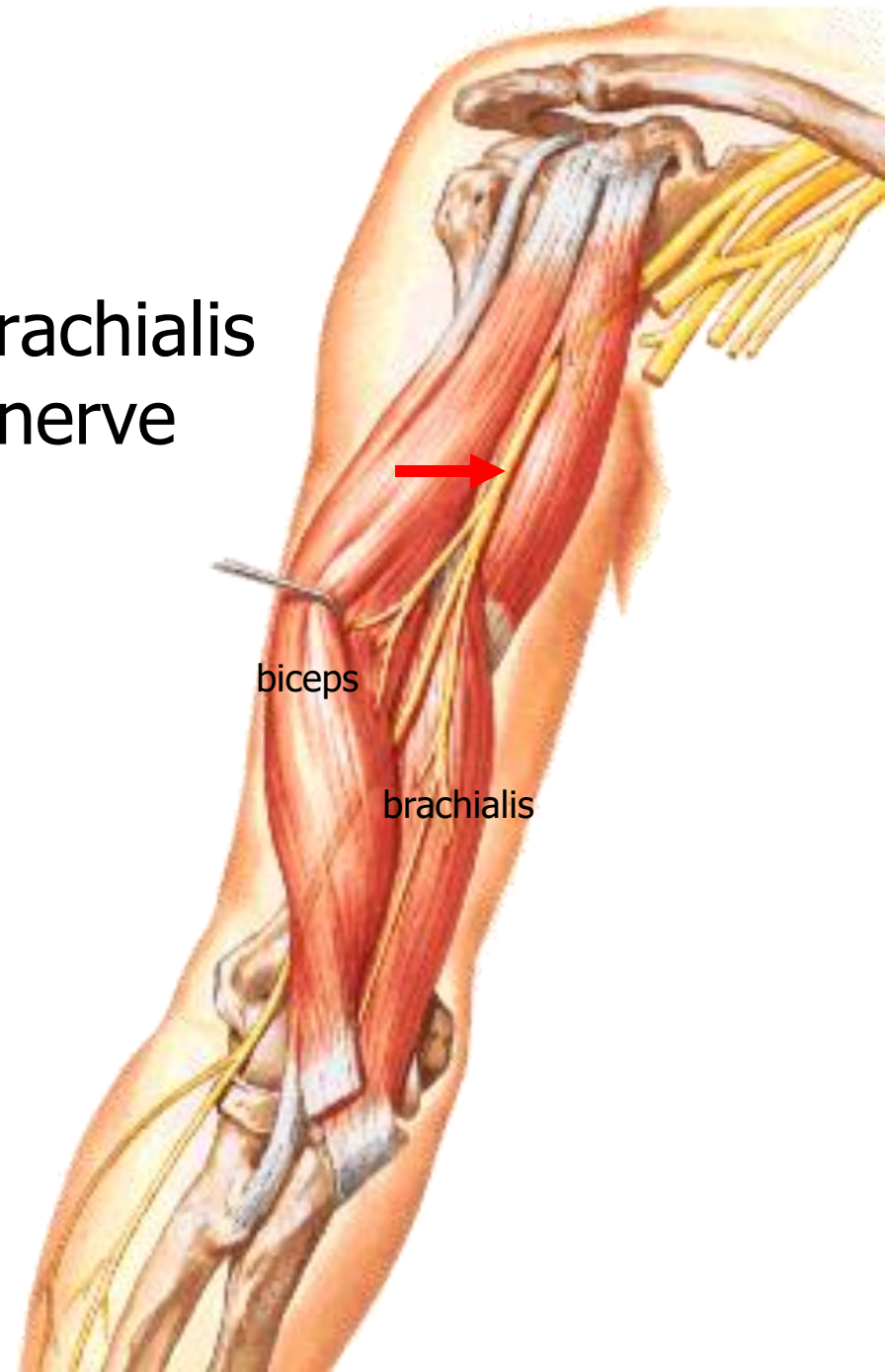
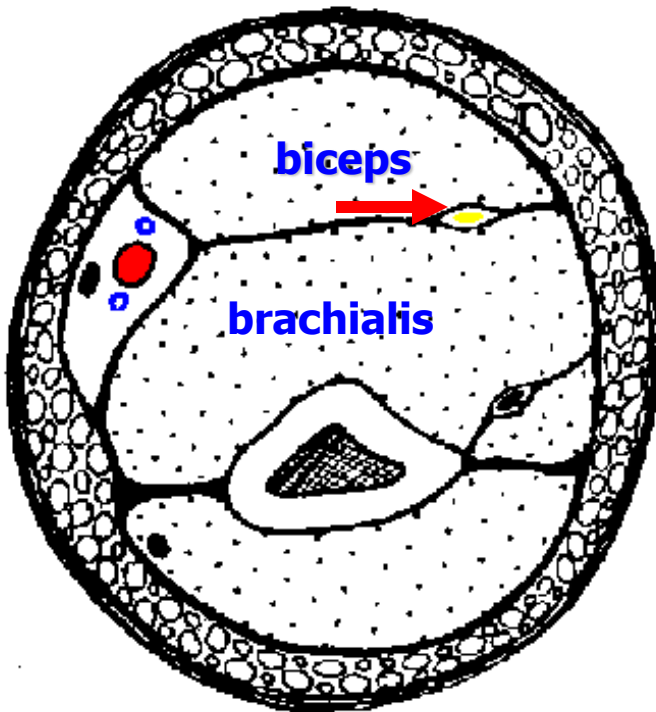
Medial head of biceps brachii

□ arises from the tip of the coracoid process of the scapula in common tendon with coracobrachialis muscle



Biceps brachii

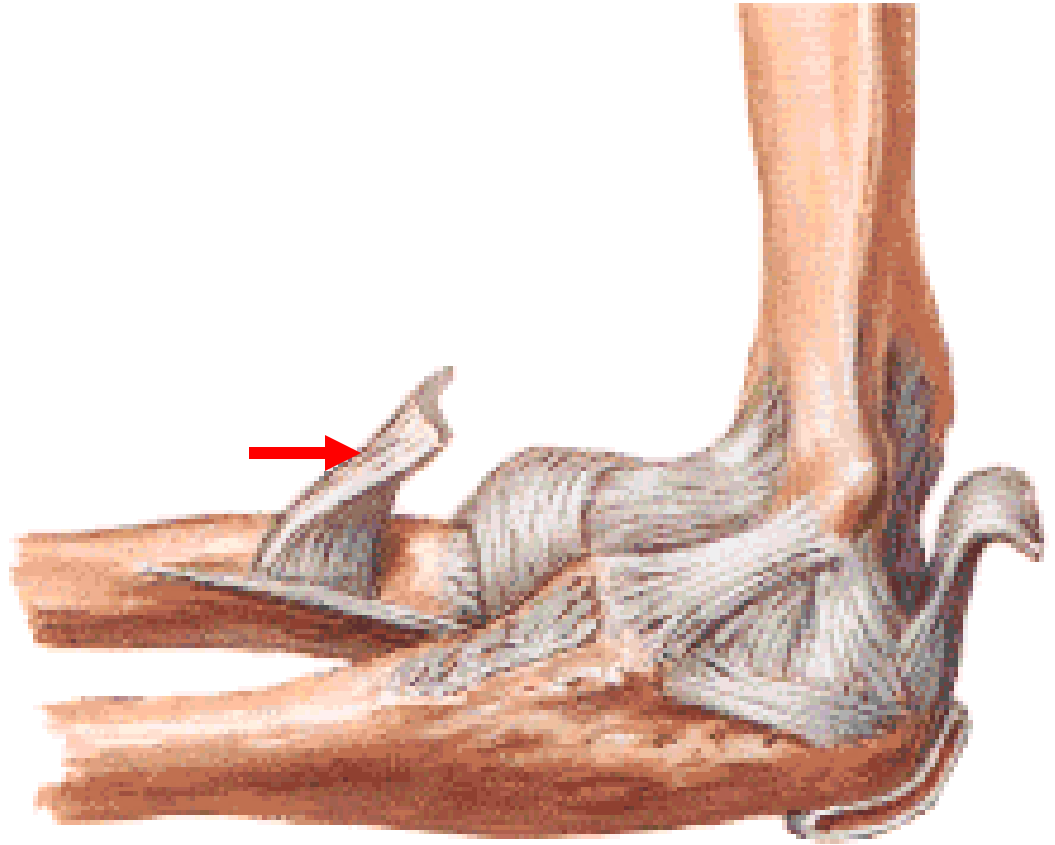
- The muscle lies in front of brachialis with the musculocutaneous nerve sandwiched between them



biceps brachii

□ Insertion

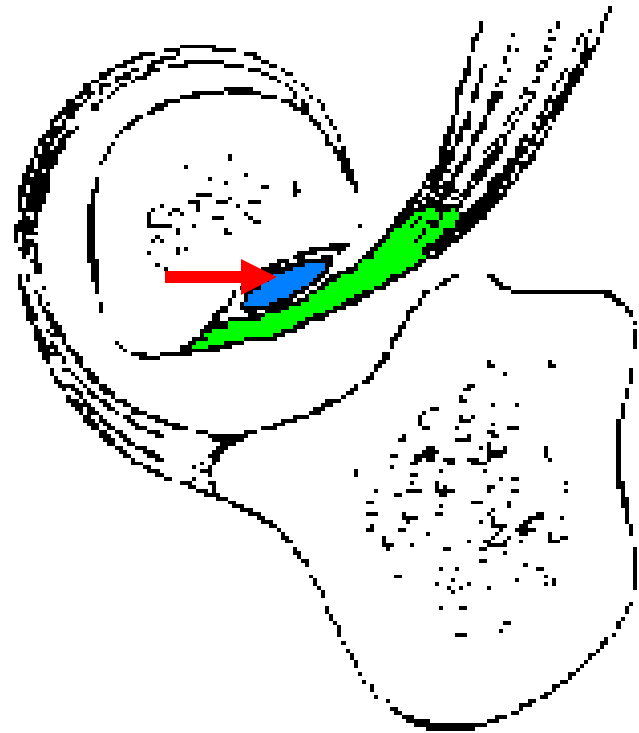
into the radial
tuberosity



Insertion of biceps brachii

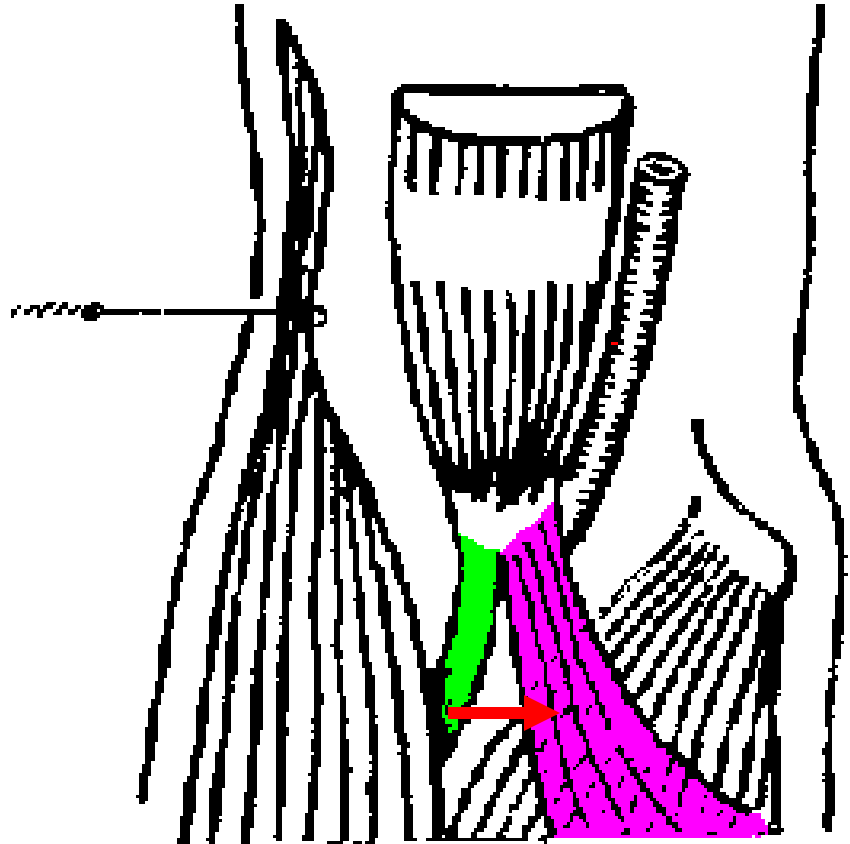
□ A bursa

is present between the tendon and radial tuberosity



Bicipital aponeurosis

- An extension from the tendon of biceps brachii
- passes medially to blend with the deep fascia of the forearm



biceps brachii

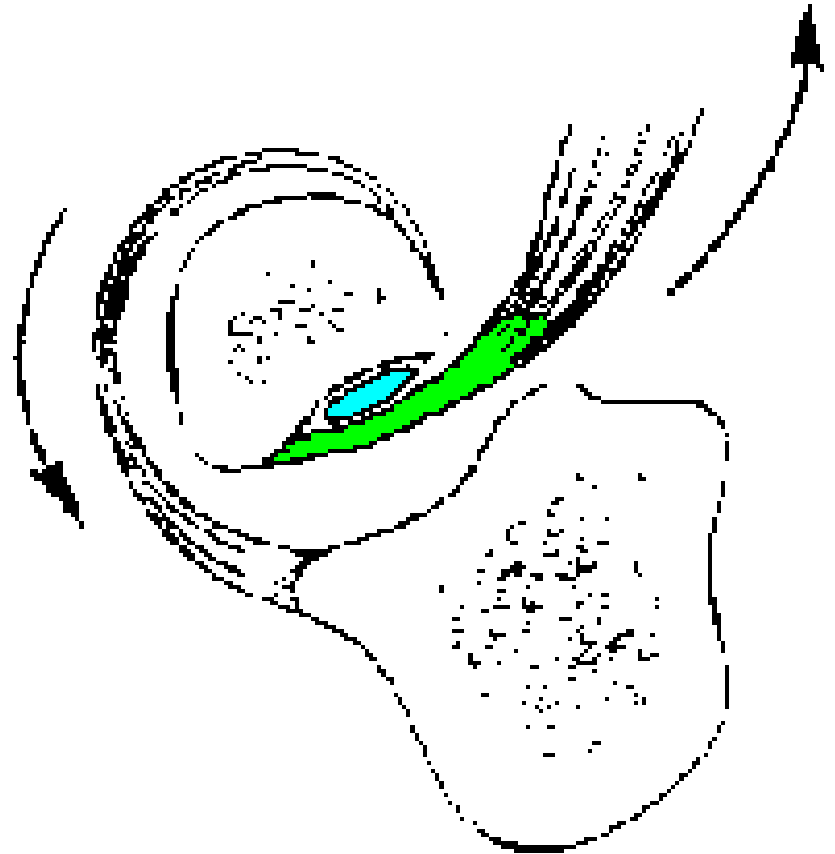
□ Action

Has TWO main actions &
ONE minor action

MAIN

1) because of its insertion
onto the posterior aspect of
the radial tuberosity

It is a powerful supinator of
the forearm

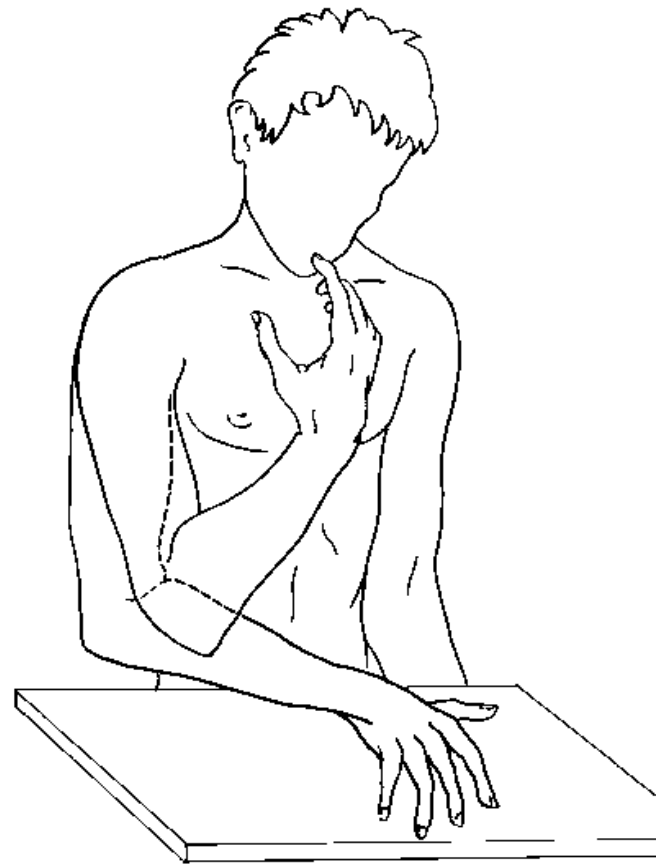


biceps brachii

□ Action

MAIN

2) It is a flexor of the forearm



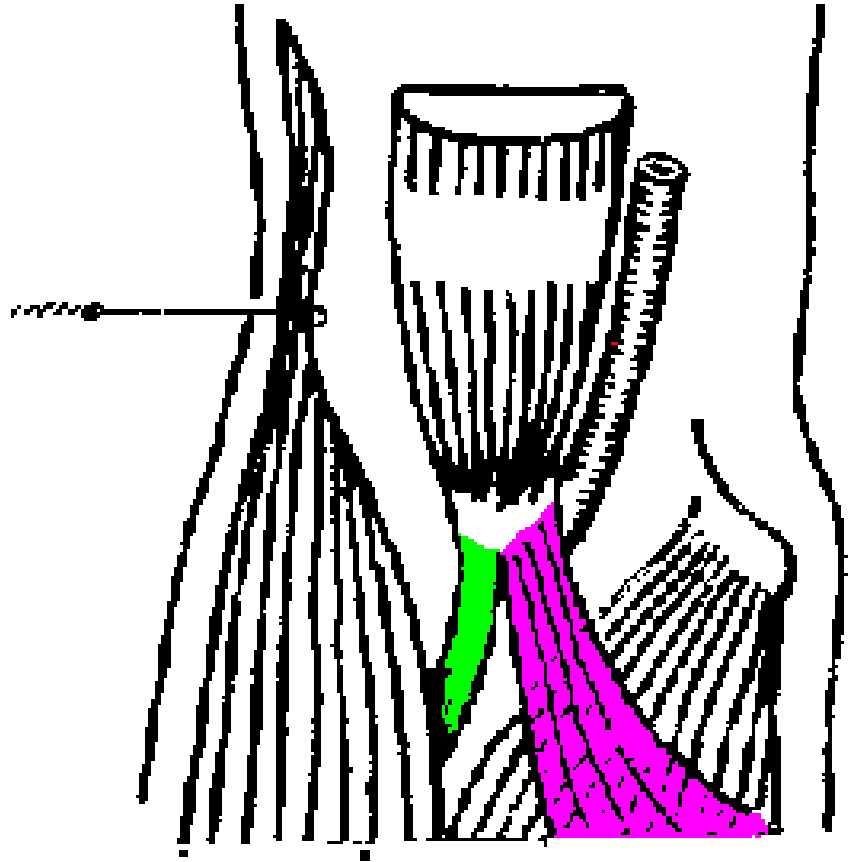
Action of biceps brachii

□ Third action is

its long head arising from the supraglenoid tubercle

a) aids in flexion of the shoulder joint

b) is an important factor in the stability of the abducted shoulder



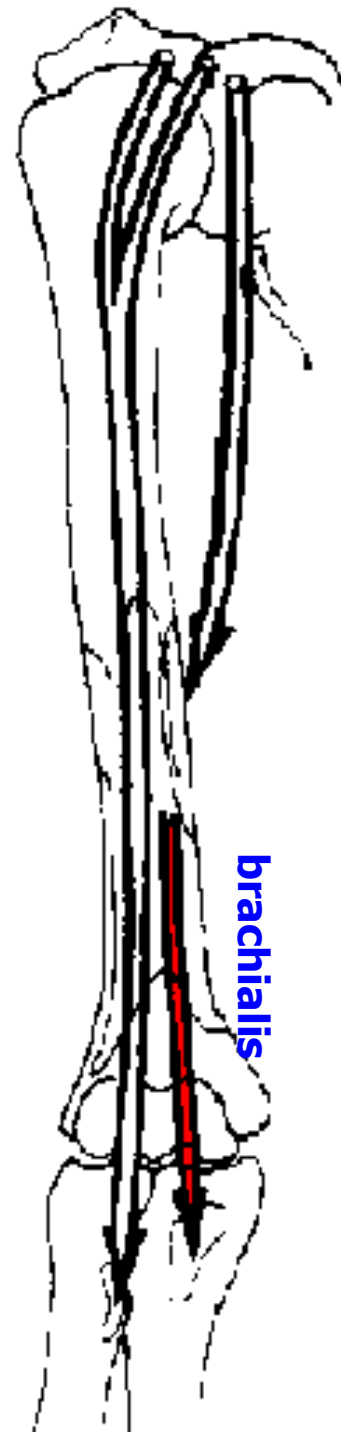
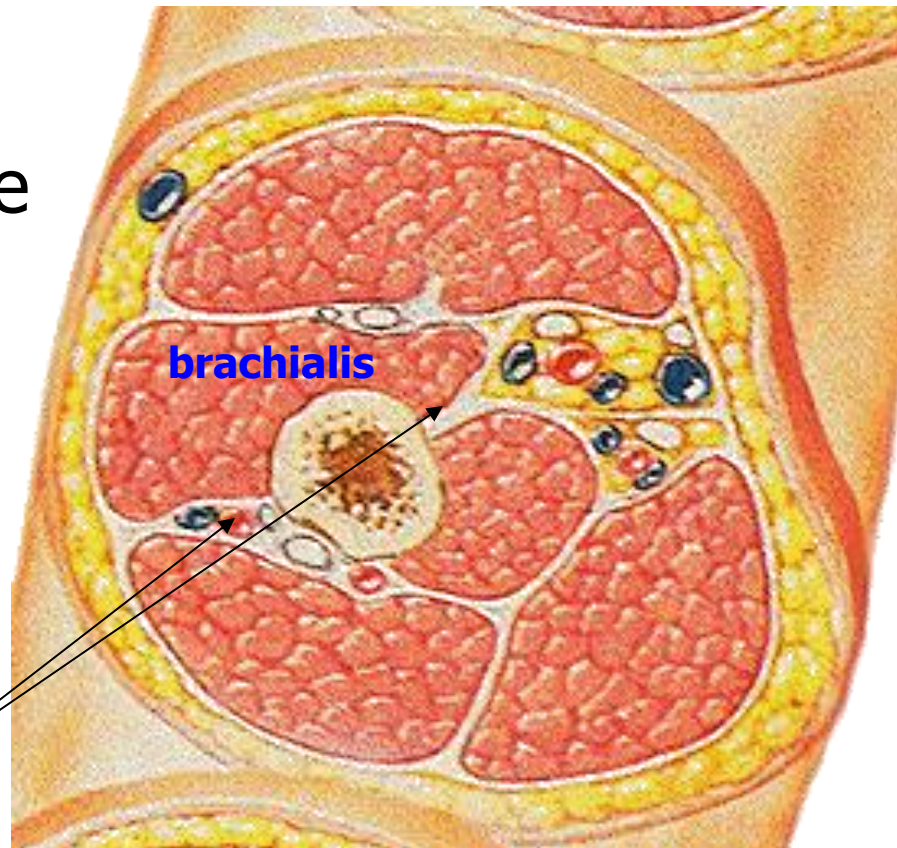
2) Brachialis origin

□ Origin

from:

- front of the lower half of the humerus
- intermuscular septa

Medial & lateral
intermuscular septa

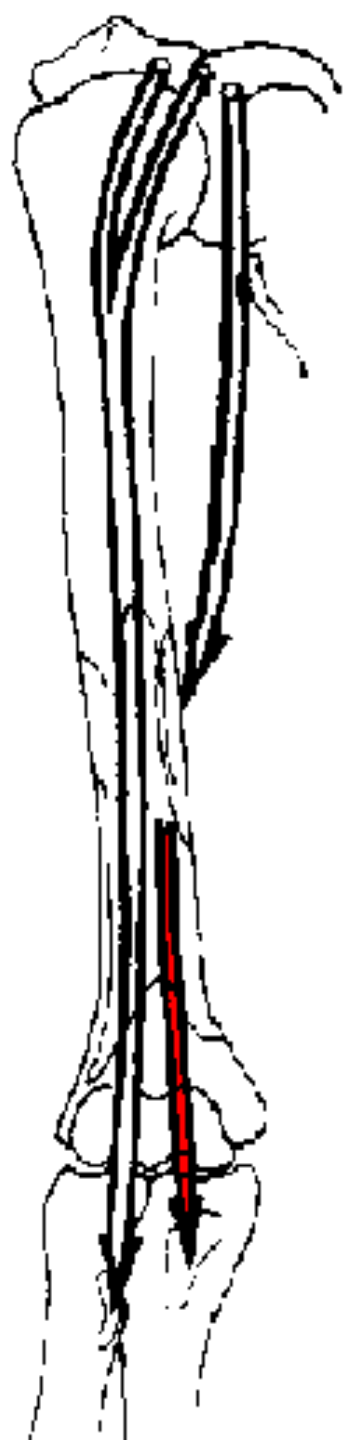


Brachialis insertion

□ Insertion

Into

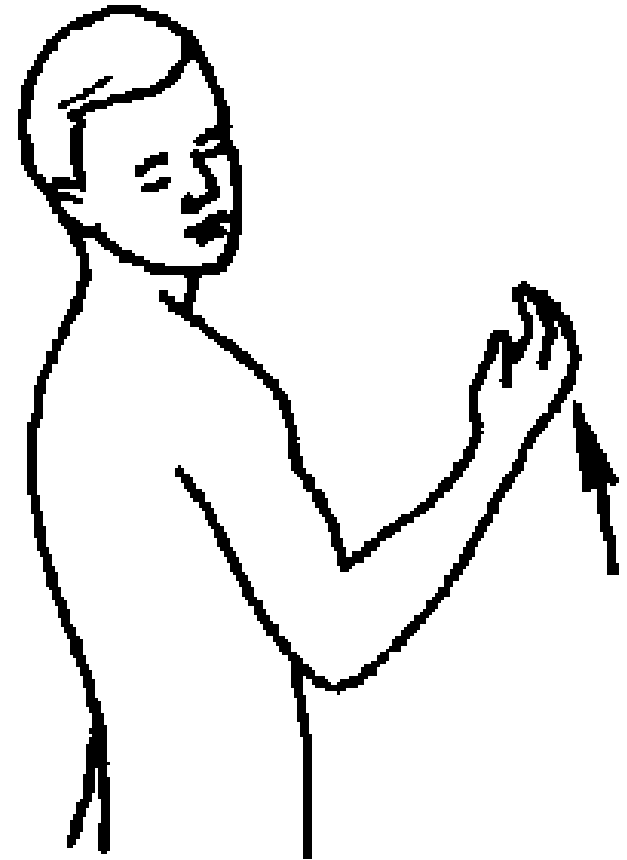
- the coronoid process
- and tuberosity of the ulna.



Brachialis action

□ Action

- Passing in front of the elbow joint
- So it is a powerful flexor of the elbow



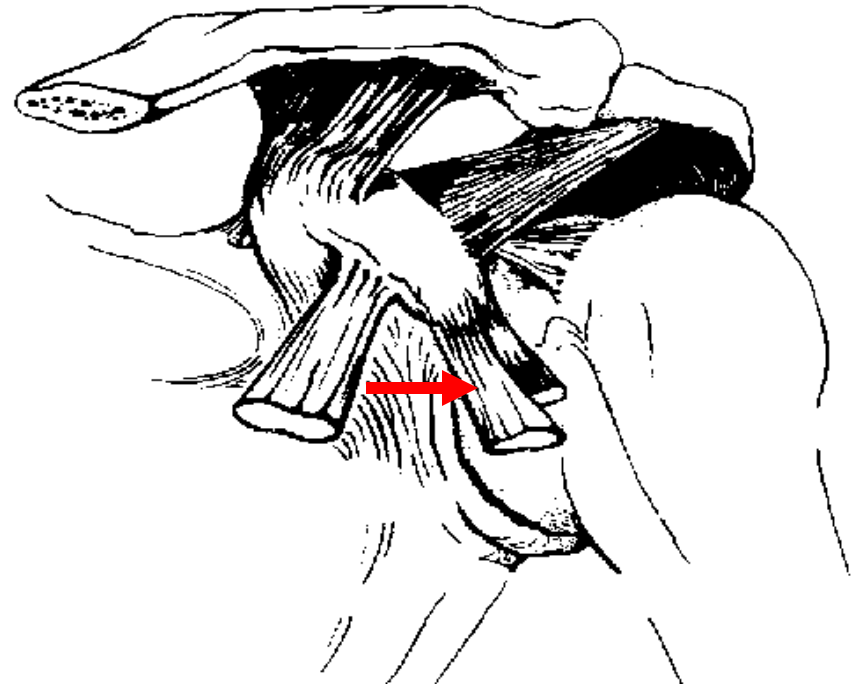
NOTE:

Brachialis has dual innervation (double); Musculocutaneous & radial

3) Coracobrachialis

Origin

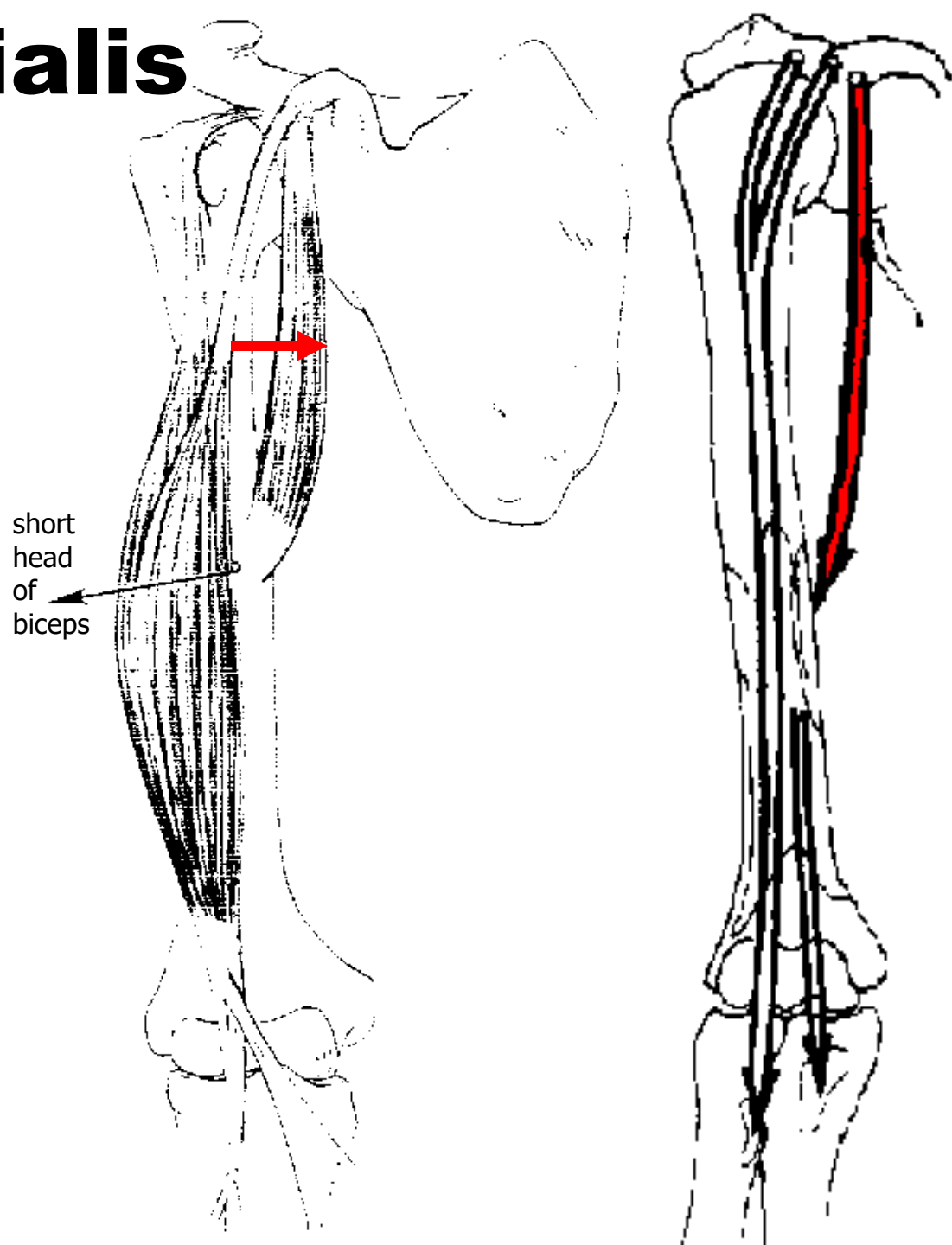
- Is a small muscle
- arises from the tip of the coracoid process in common with the short head of biceps brachii



Coracobrachialis

insertion

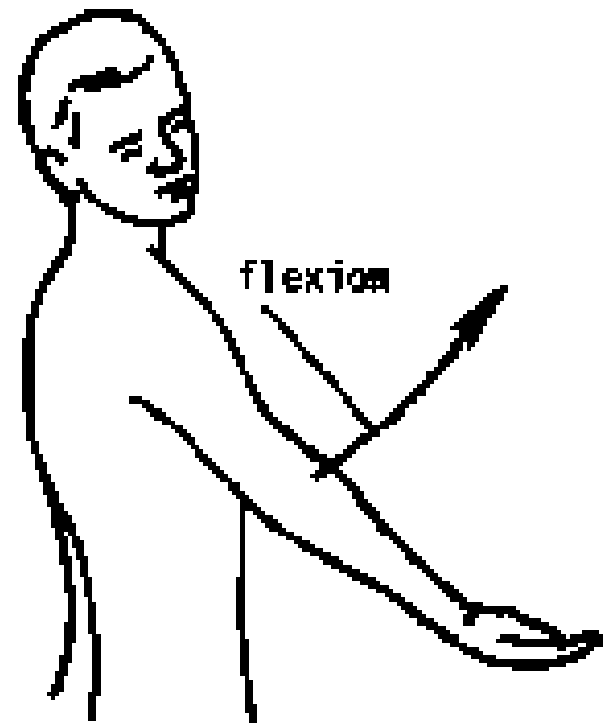
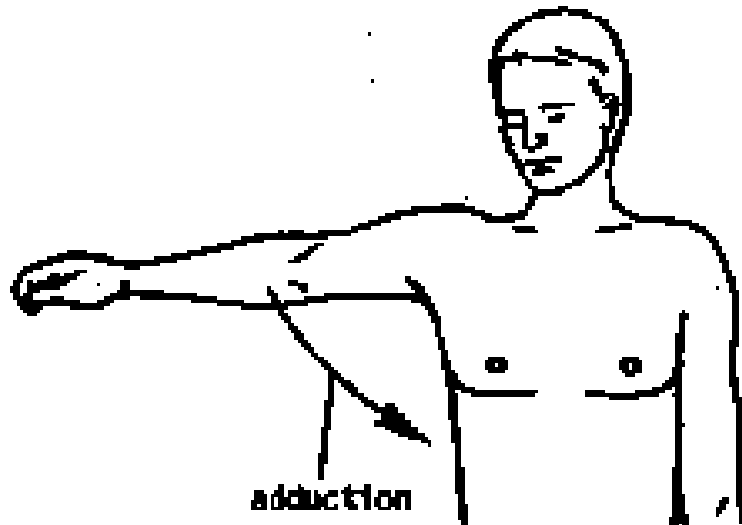
- into the middle point of medial border of the humerus.
- Adjacent to insertion of deltoid at the middle point of lateral border



Coracobrachialis

action

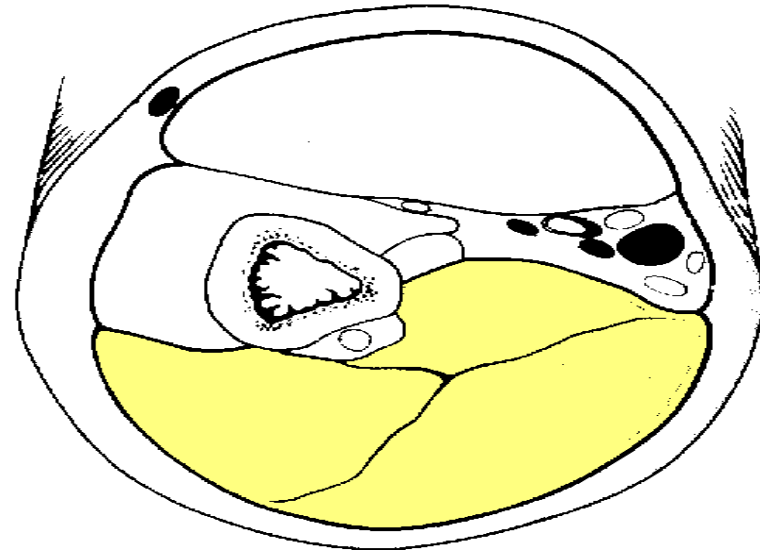
- It is a weak flexor and adductor of the shoulder



POSTERIOR COMPARTMENT

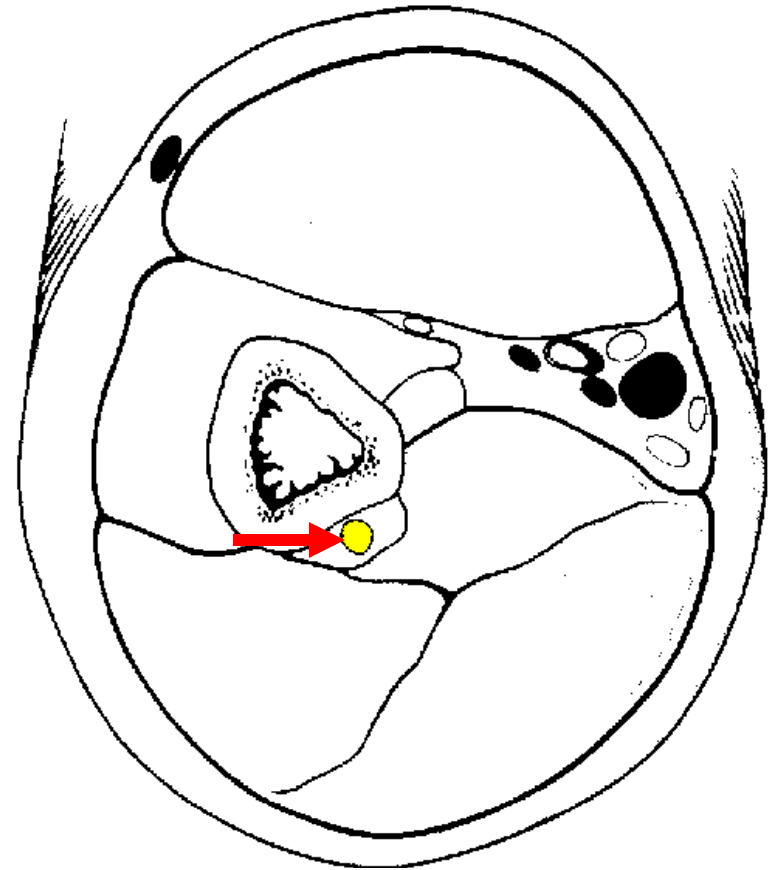
B. Muscles of Back of arm

Muscle	Origin	Insertion	Nerve supply	Action.
Triceps Brachii	a. Long head: infra-glenoid tubercle (extra-capsular) b. Lateral head: Back of shaft of humerus above spiral groove. c. Medial head: Back of shaft of humerus below spiral groove.	Superior surface of olecranon process of ulna.	Radial nerve (long head is supplied in axilla while lateral & medial heads are supplied in spiral groove).	Main extensor of elbow.



Posterior compartment of the arm

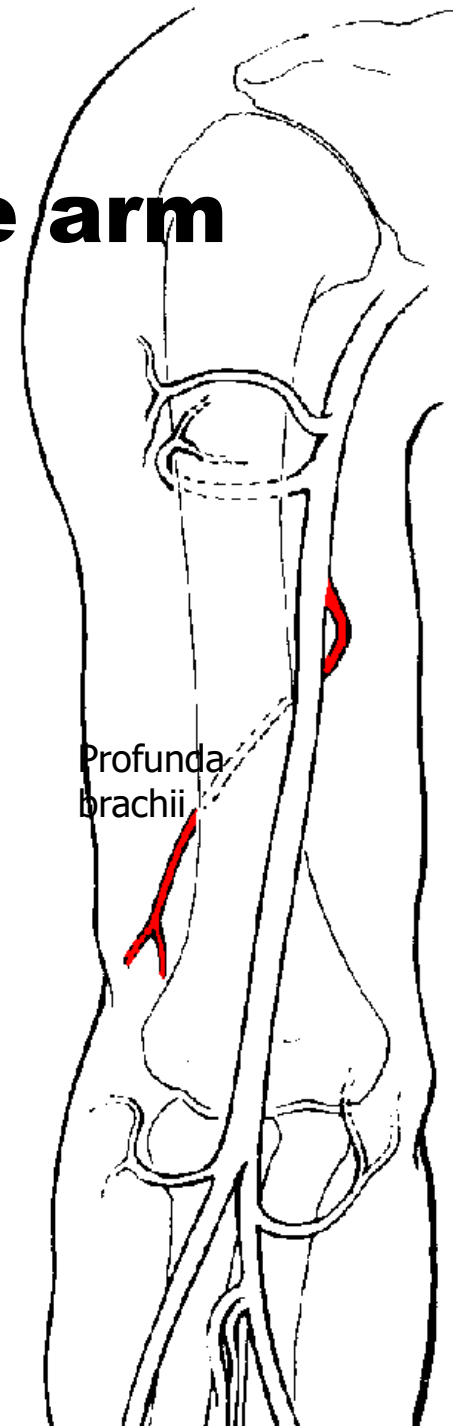
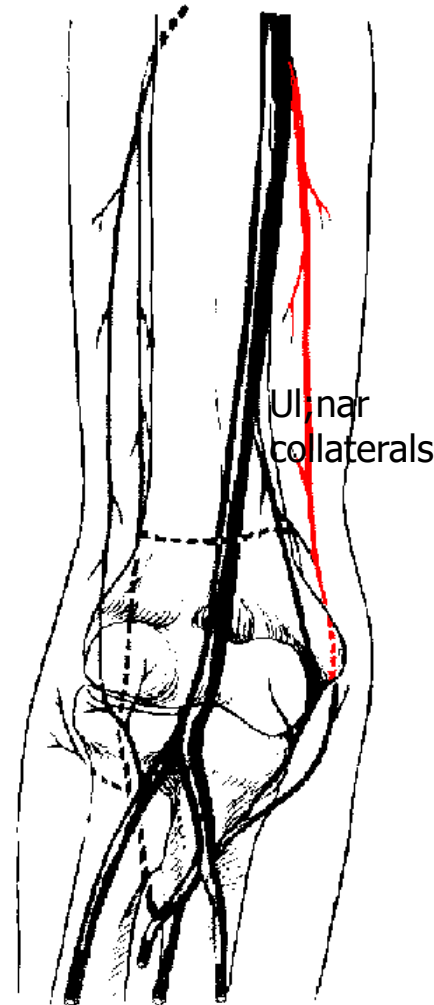
- the nerve of the compartment is the RADIAL NERVE



Posterior compartment of the arm

- The arteries of the compartment are
 - profunda brachii
 - ulnar collateral arteries

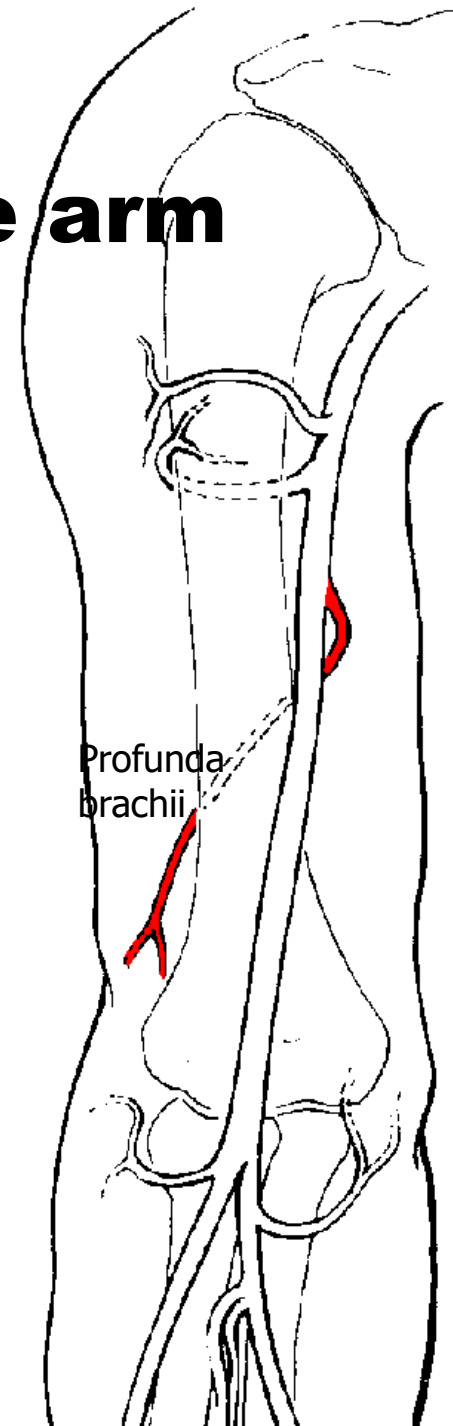
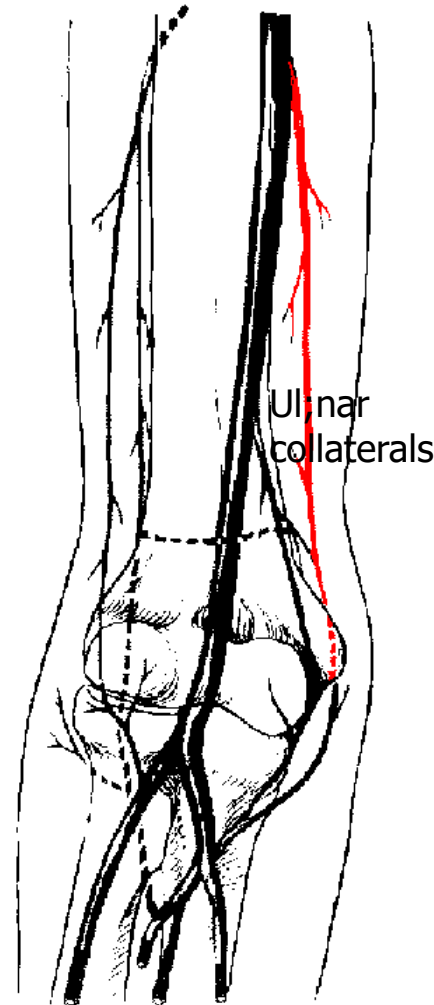
They are branches of the brachial artery



Posterior compartment of the arm

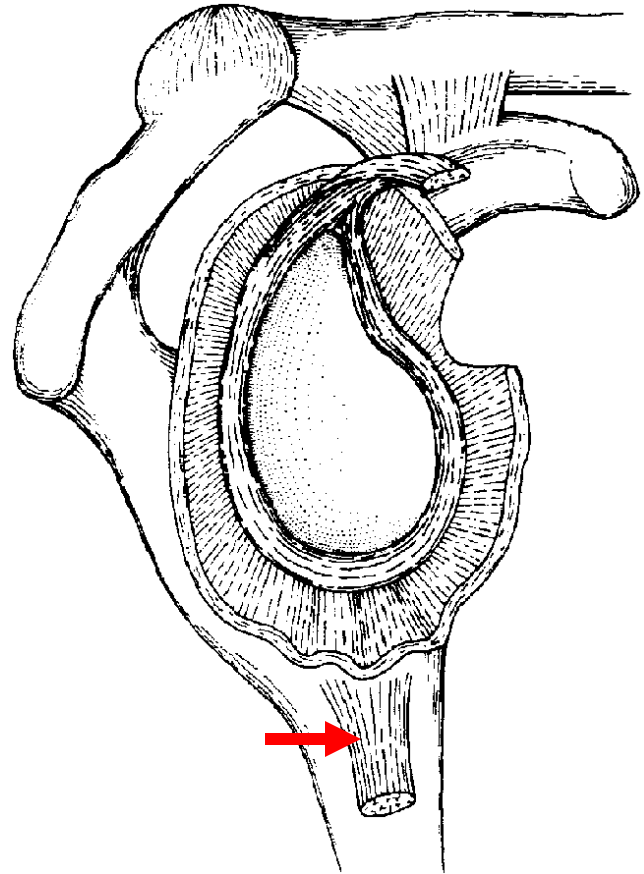
- The arteries of the compartment are
 - profunda brachii
 - ulnar collateral arteries

They are branches of the brachial artery



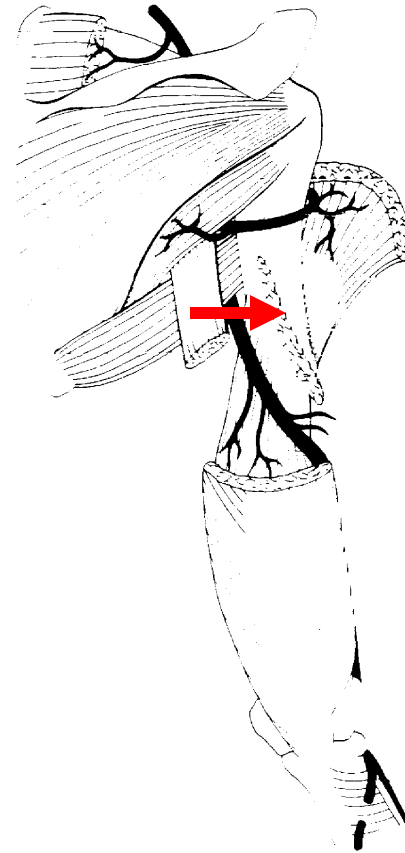
Long head of triceps

- arises from the infraglenoid tubercle of the scapula



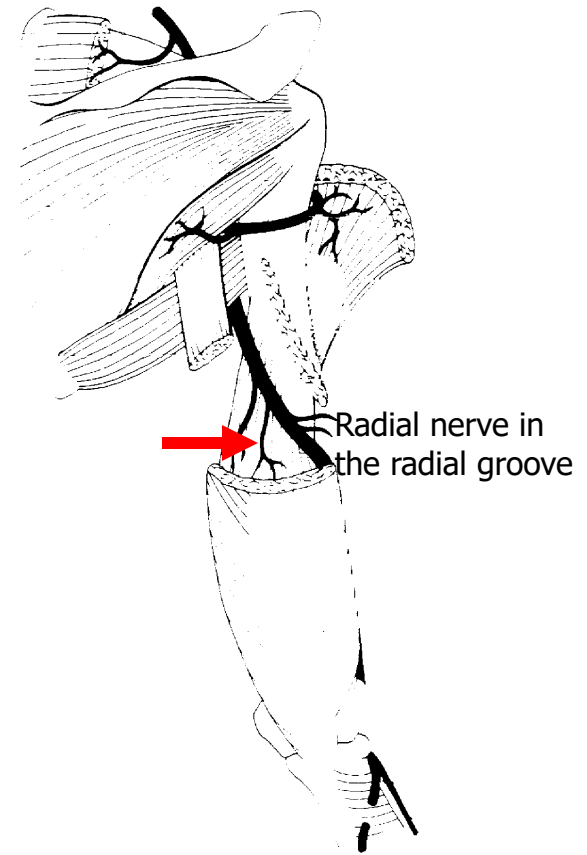
Lateral head of triceps

- arises from the posterior surface of the humerus superior to the radial groove



Medial head of triceps

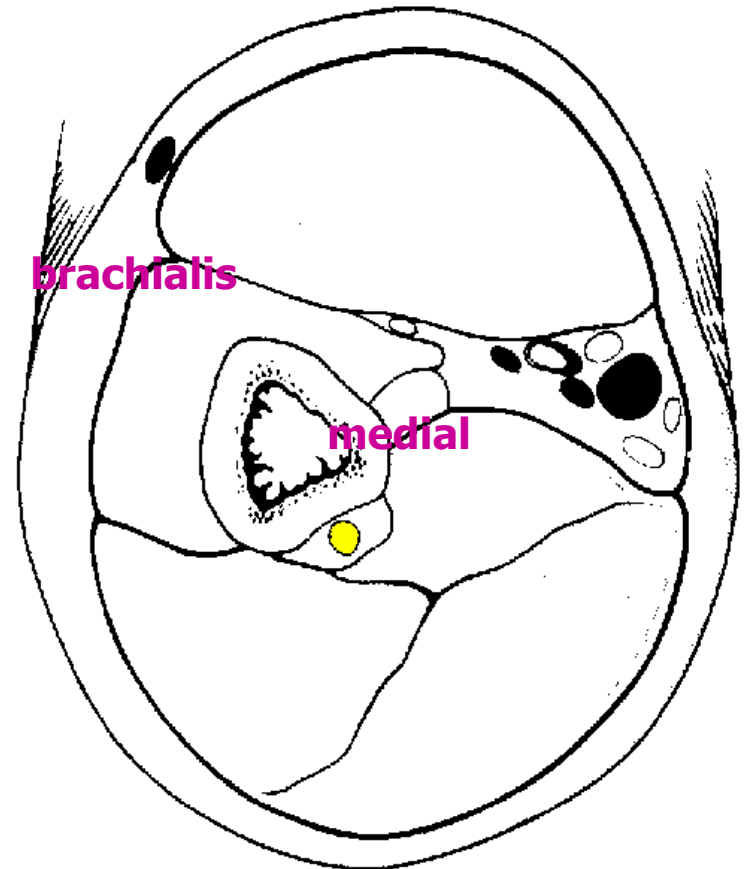
- Arises from the posterior surface of the humerus *inferior* to the radial groove



Triceps heads

□ the medial head is comparable to brachialis

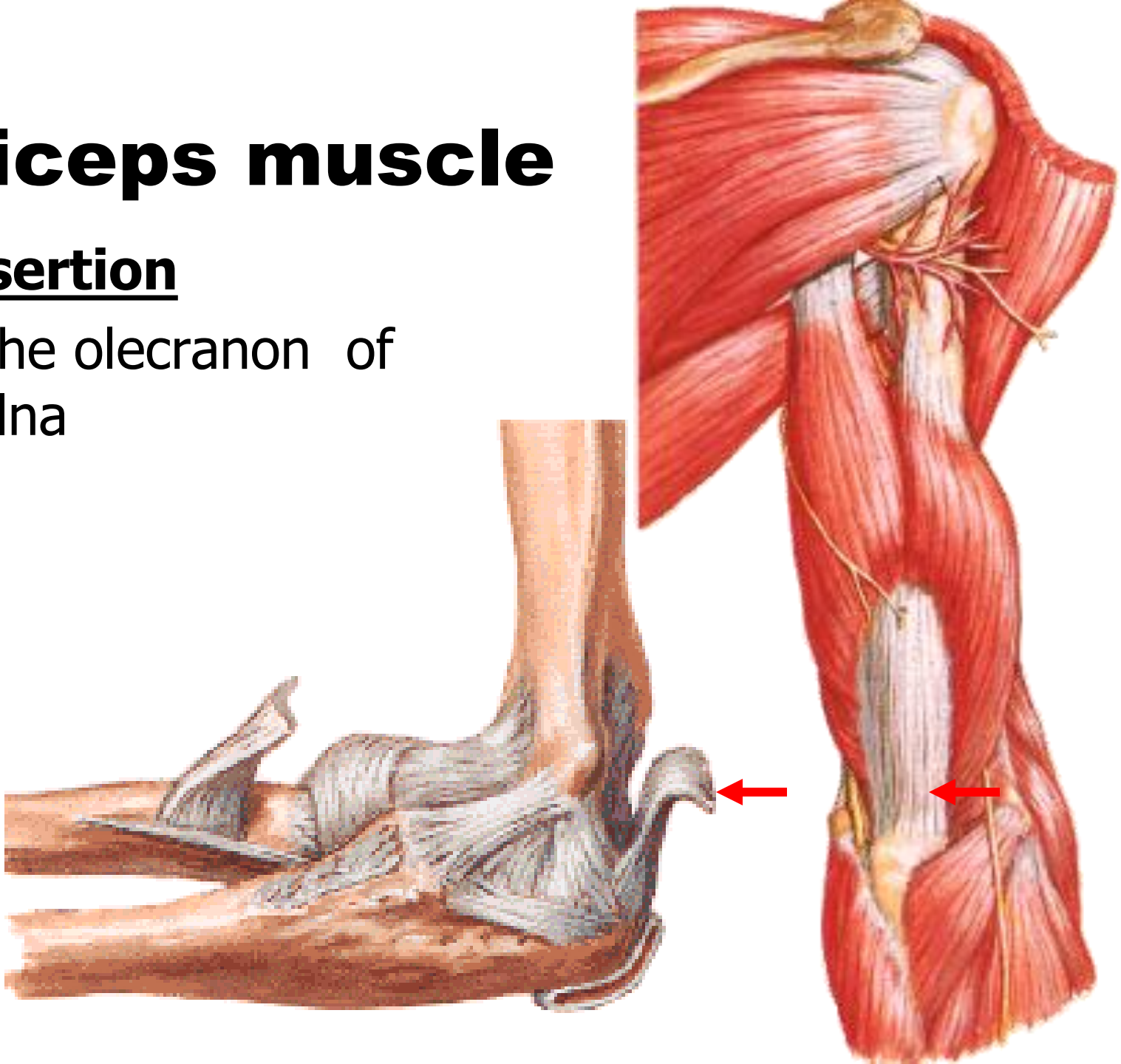
- Lies in a deeper plane than other heads
- takes origin from the lower part of the shaft and the intermuscular septum



Triceps muscle

□ Insertion

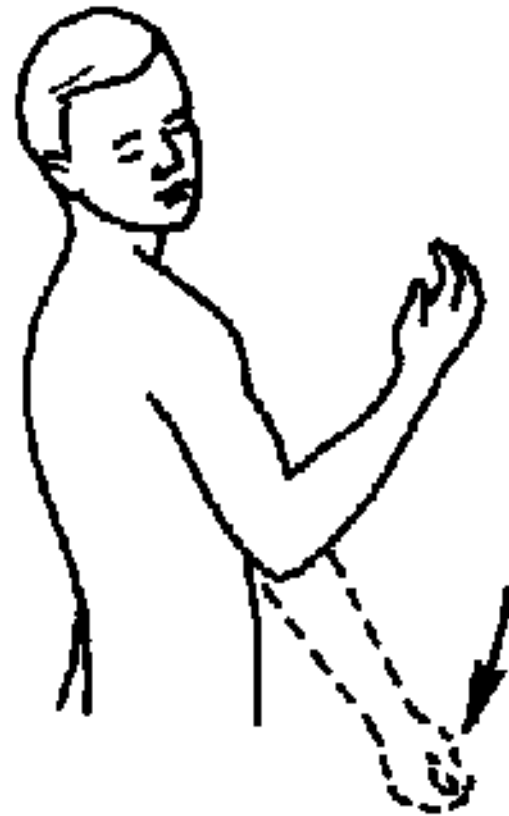
into the olecranon of
the ulna



Triceps action

action

- 1) The muscle is an *extensor* of the elbow



Triceps action

2) its long head arising from the infraglenoid tubercle

a) aids in extending the shoulder joint

b) is an important factor in the stability of the abducted shoulder



Bursae related to triceps tendon

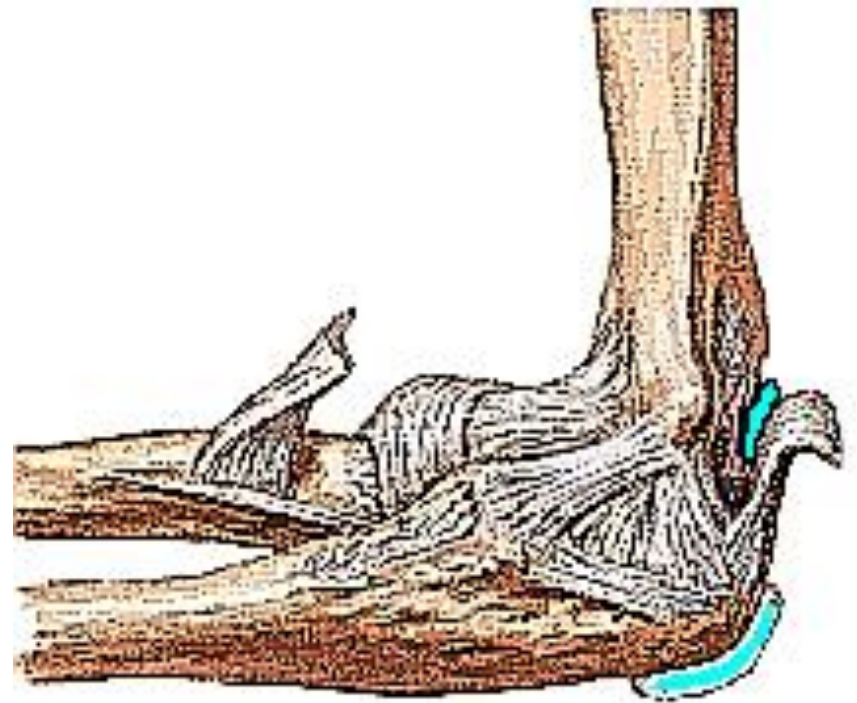
□ It related to TWO bursae

1) Subcutaneous bursa

□ located in the
subcutaneous tissue

2) Subtendinous olecranon
bursa

□ located between the
tendon and olecranon



Student's elbow

- The subcutaneous bursa is liable to be inflamed by repeated friction this is called "student's elbow "

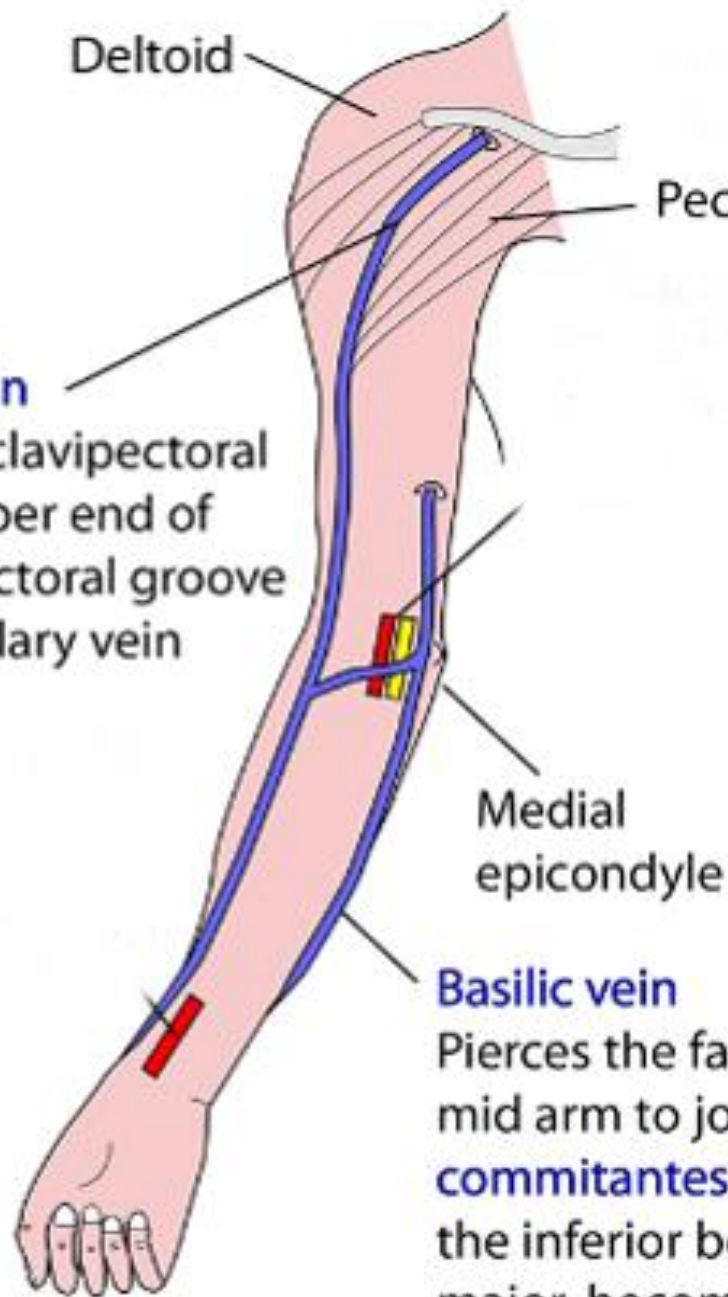


ARTERIES

SUPERFICIAL VEINS IN UPPER LIMB

Cephalic vein

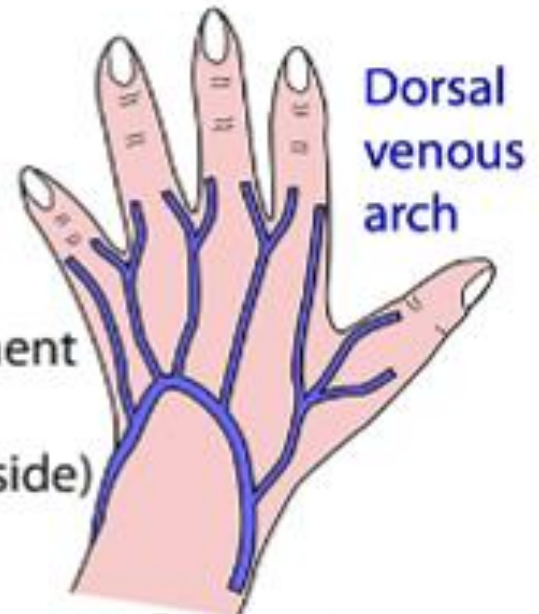
Pierces the clavipectoral fascia at upper end of the deltopectoral groove to enter axillary vein



Basilic vein

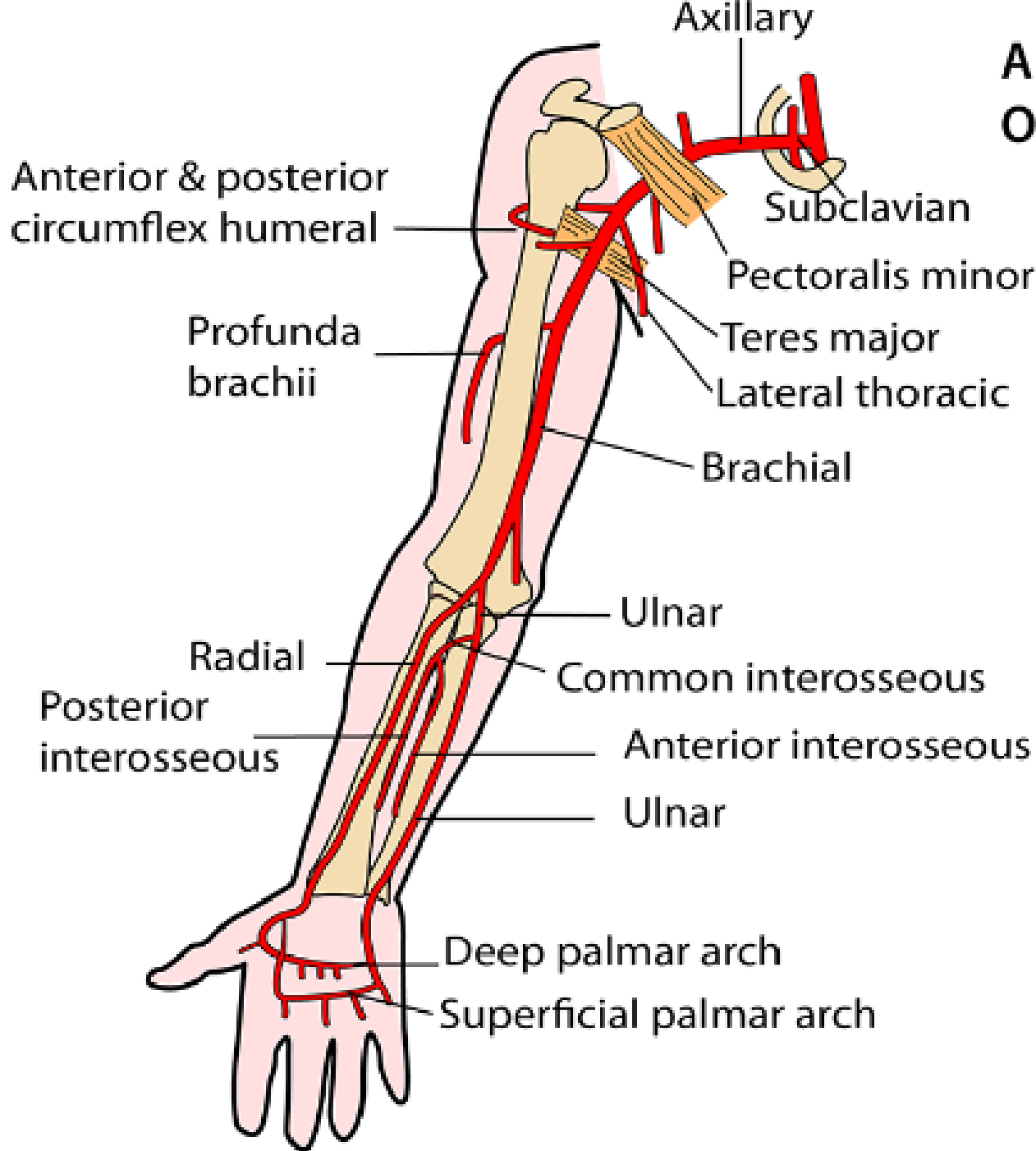
Pierces the fascia in the medial mid arm to join the **venae comitantes** which together, at the inferior border of **teres major**, become the **axillary vein**

Commencement of the **basilic vein** (medial side)



Commencement of the **cephalic vein** (lateral side)

ARTERIAL SUPPLY OF UPPER LIMB



Other branches:

Muscular

Nutrient

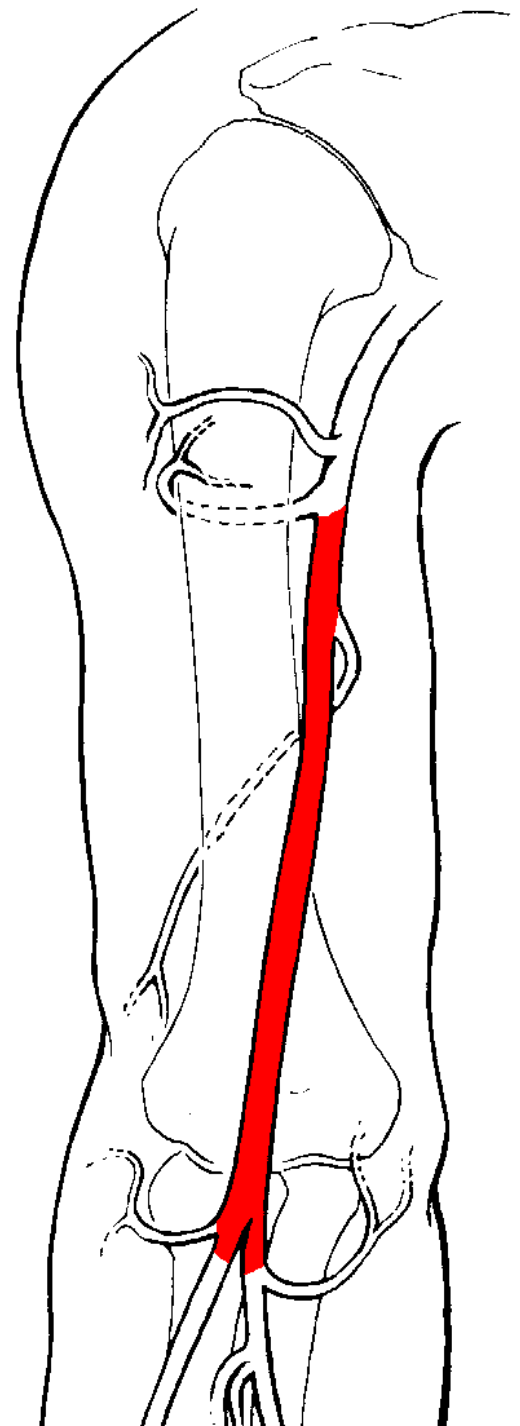
Ulnar collateral

All arteries
accompanied by
venae comitantes

B) Brachial artery

□ Beginning

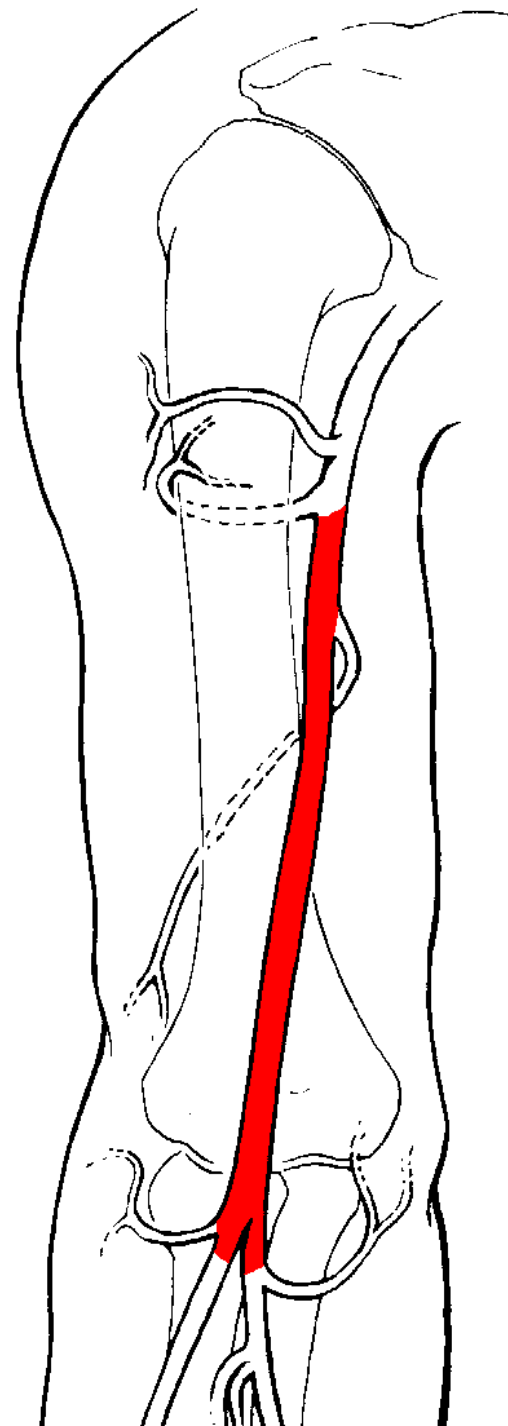
is the continuation of the axillary artery at lower border of teres major m.



Brachial artery

□ Ends

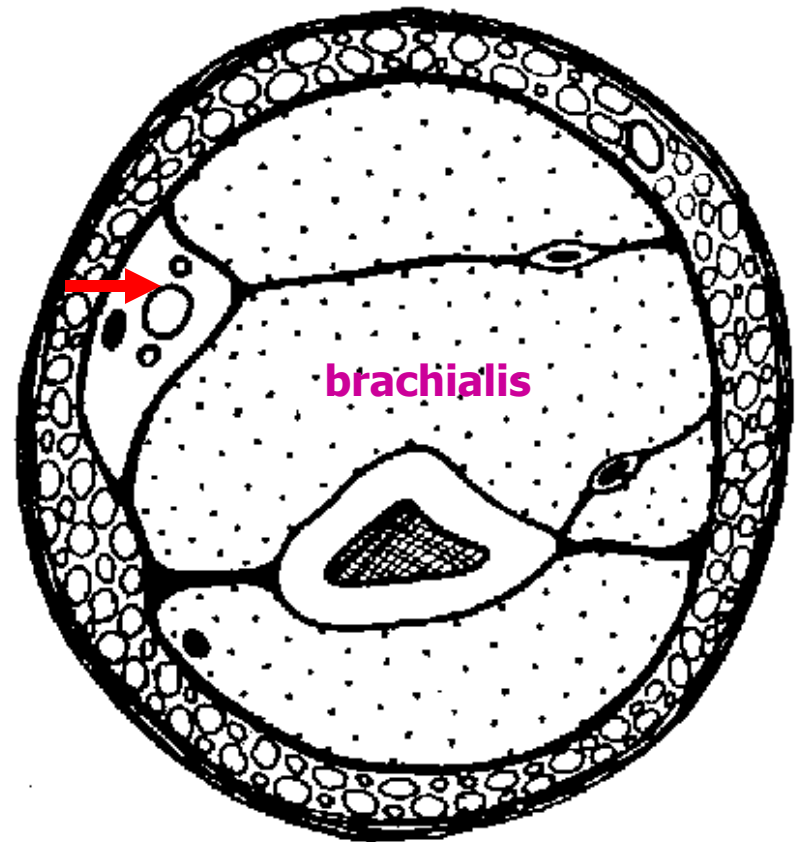
- by dividing into radial and ulnar arteries
- At the cubital fossa in front of the elbow joint
- opposite the neck of the radius



Brachial artery

□ Course

It runs on the medial border of brachialis muscle

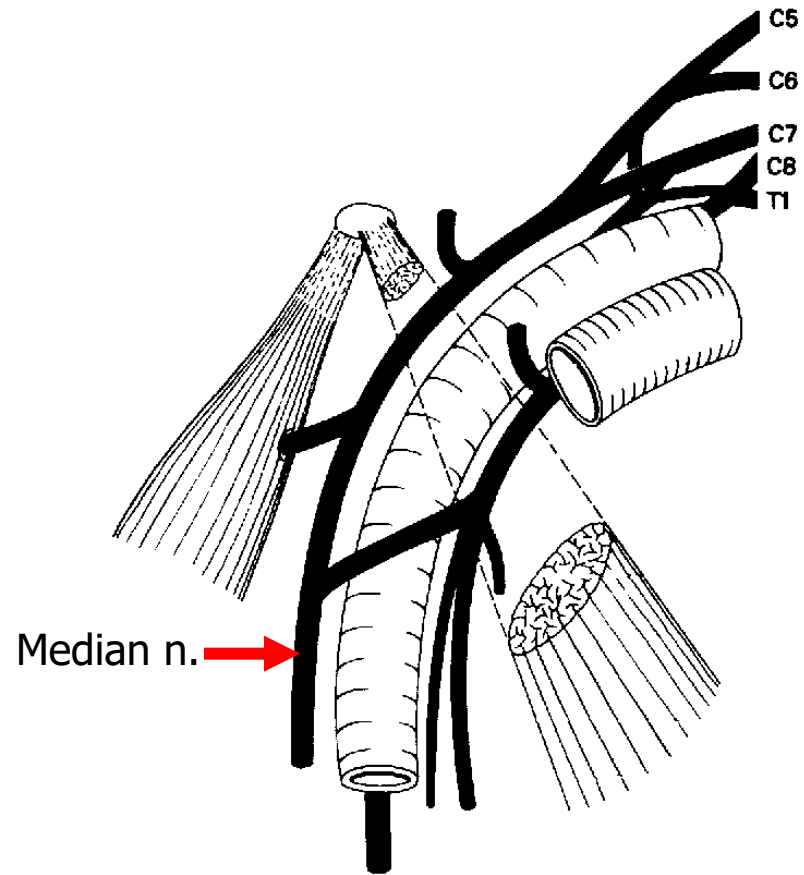


Brachial artery

□ Relation to median nerve

1) At axilla

The median nerve is formed by the union of the medial and lateral roots lateral to the axillary artery

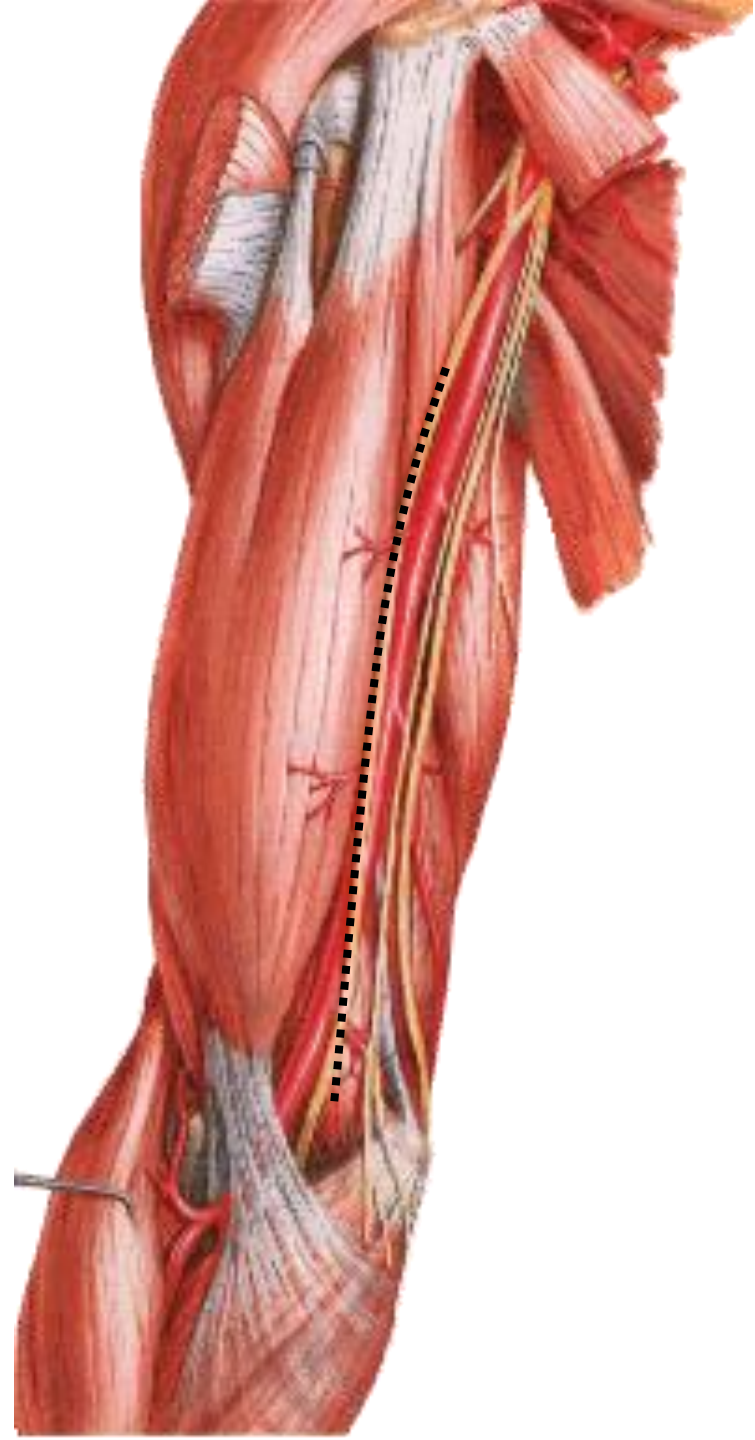


Brachial artery

□ Relation to median nerve

2) At arm

The median nerve crosses anterior to the brachial artery in the middle of the arm

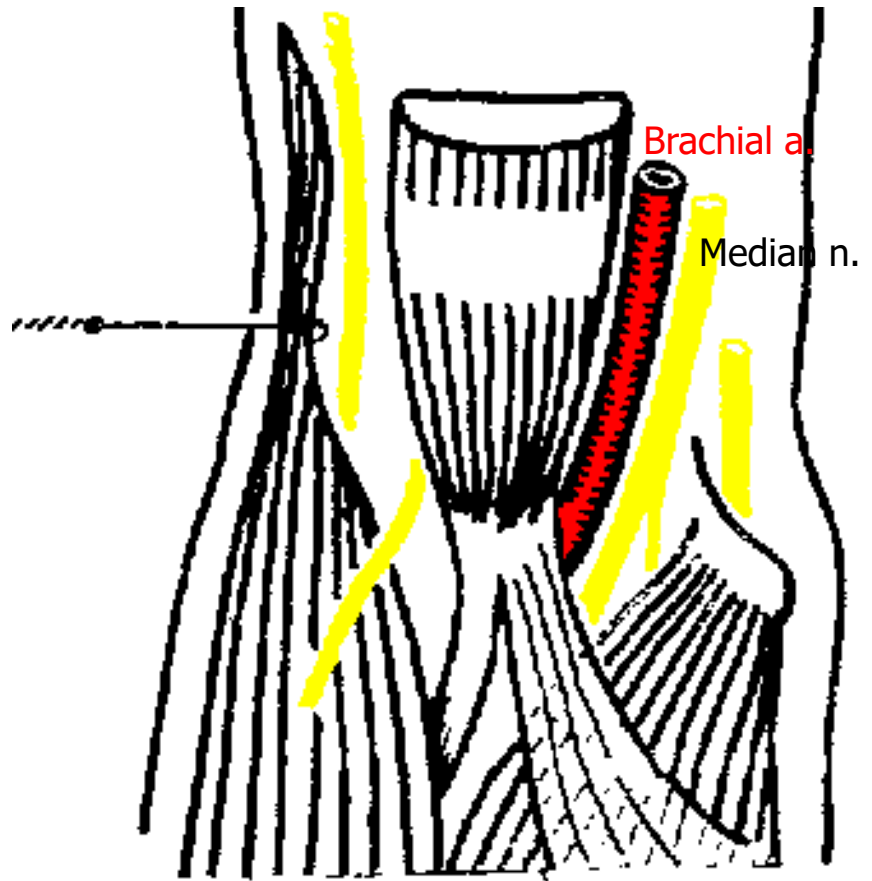


Brachial artery

□ Relation to median nerve

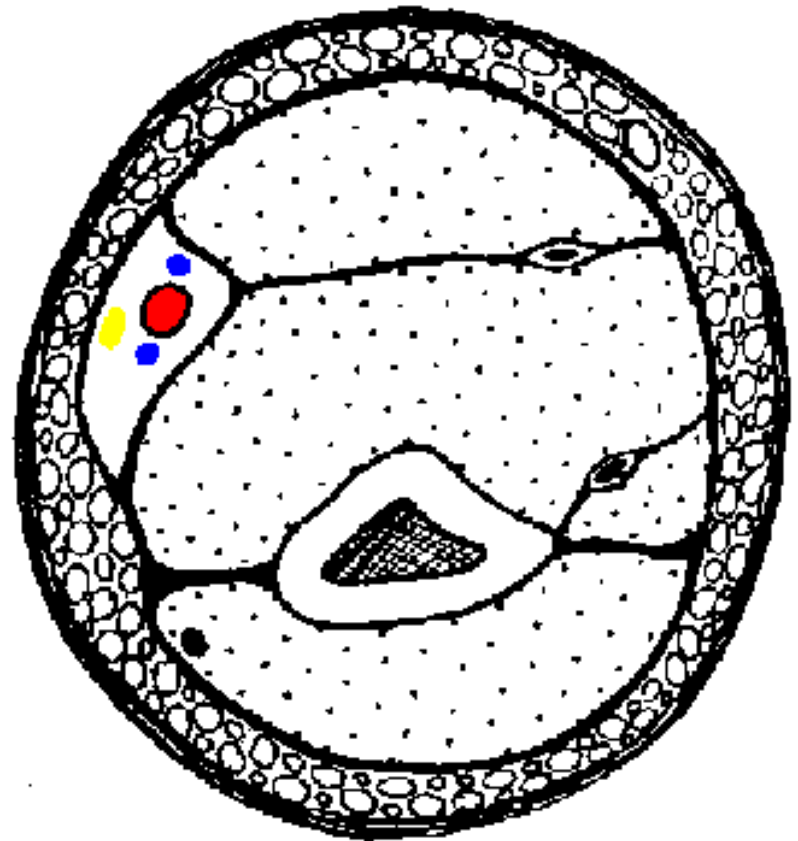
3) At the elbow

the median nerve lies
medial to the brachial artery



Brachial artery

- The artery is accompanied by **venae comitantes** (brachial veins)
- it **superficial** throughout its course lying just deep to the deep fascia



Brachial artery

❑ why auscultation of brachial artery is commonly used to record arterial blood pressure?

1- it is superficial , infront of elbow and its pulsations are easily felt just medial to tendon of biceps m.

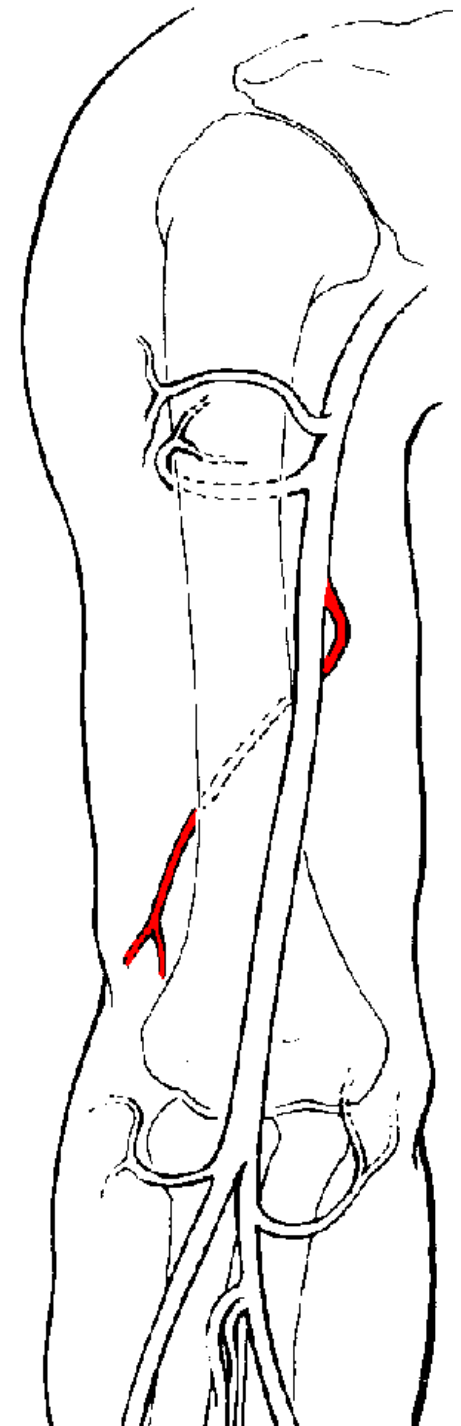
2- triangle hollow infront elbow fossa provides a flat area to place the stethoscope over artery



Branches

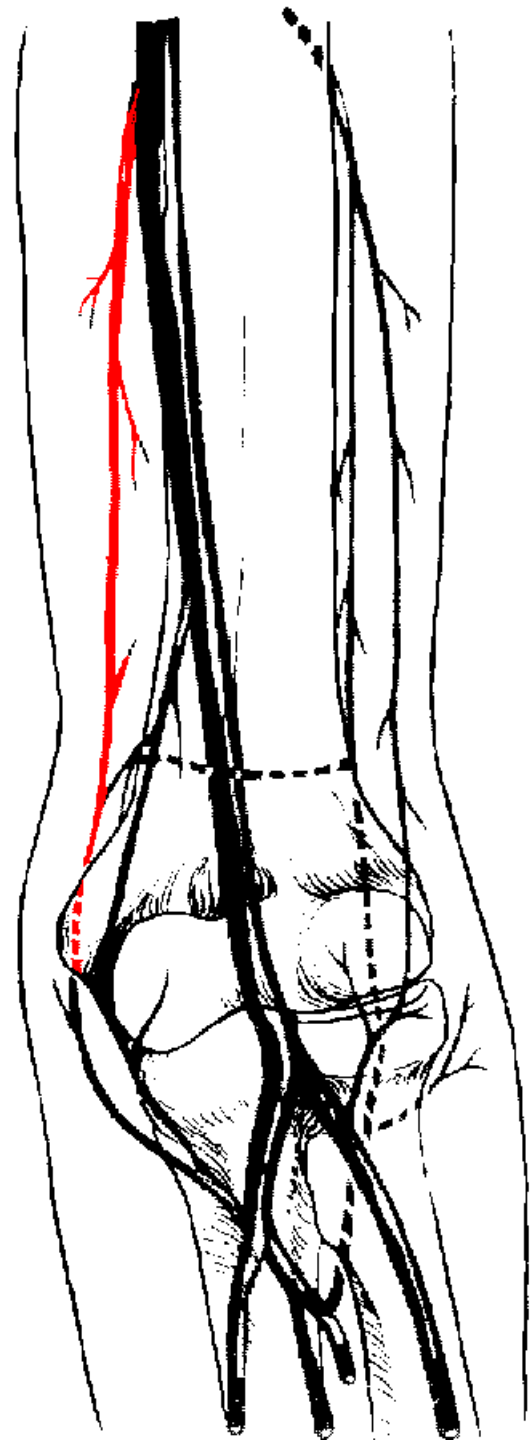
1) **Profunda brachii**

- the **first & largest** branch of the brachial artery
- Passes along the radial groove of the humerus with radial nerve
- divides into *anterior and posterior descending* branches which participate in *anastomoses around the elbow joint*



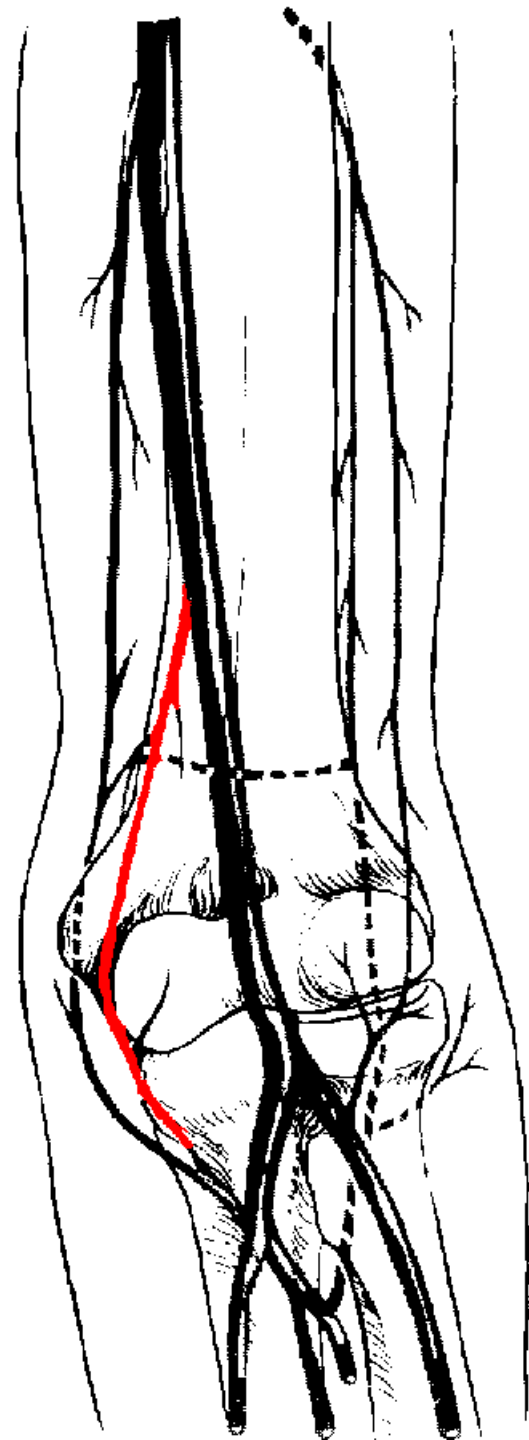
2) Superior ulnar collateral

- arises about the middle of the arm
- accompanies the ulnar nerve posterior to the medial epicondyle of the humerus.



3) Inferior ulnar collateral

- arises proximal to the elbow
- passes downwards anterior to the medial epicondyle of the humerus
- both the superior and inferior ulnar collateral arteries participate in the arterial anastomoses around the elbow joint.



Anastomosis around the elbow

□ Definition

○ arterial anastomosis around the elbow joint

○ Formed by **8** branches

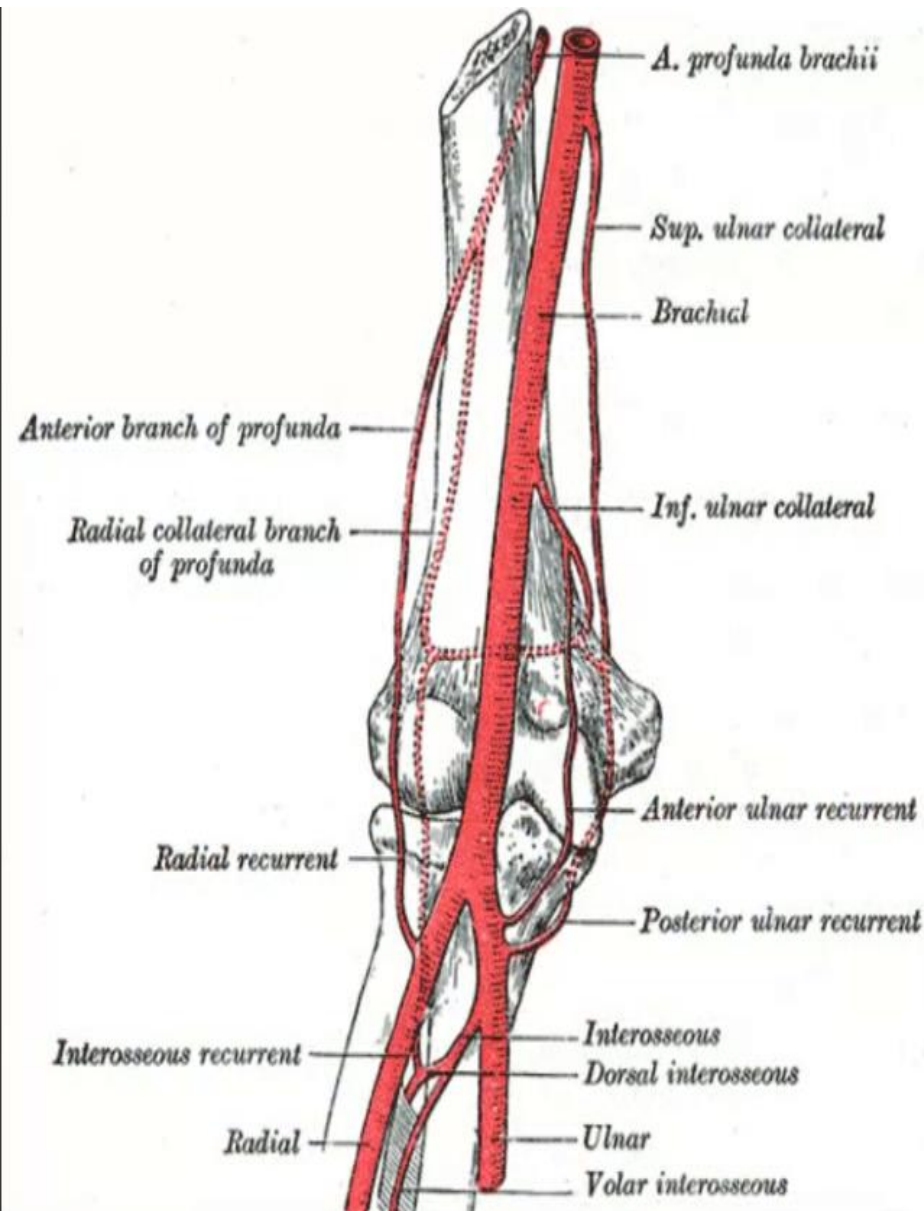
1) **Descending aa.** Which are
a) **descending branches** of the profunda brachii

b) Superior ulnar collateral a.

c) inferior ulnar collateral a.

2) **Recurrent branches** that arise from:

➤ radial a., ulnar a. & posterior interosseous a.



Anastomosis around the elbow

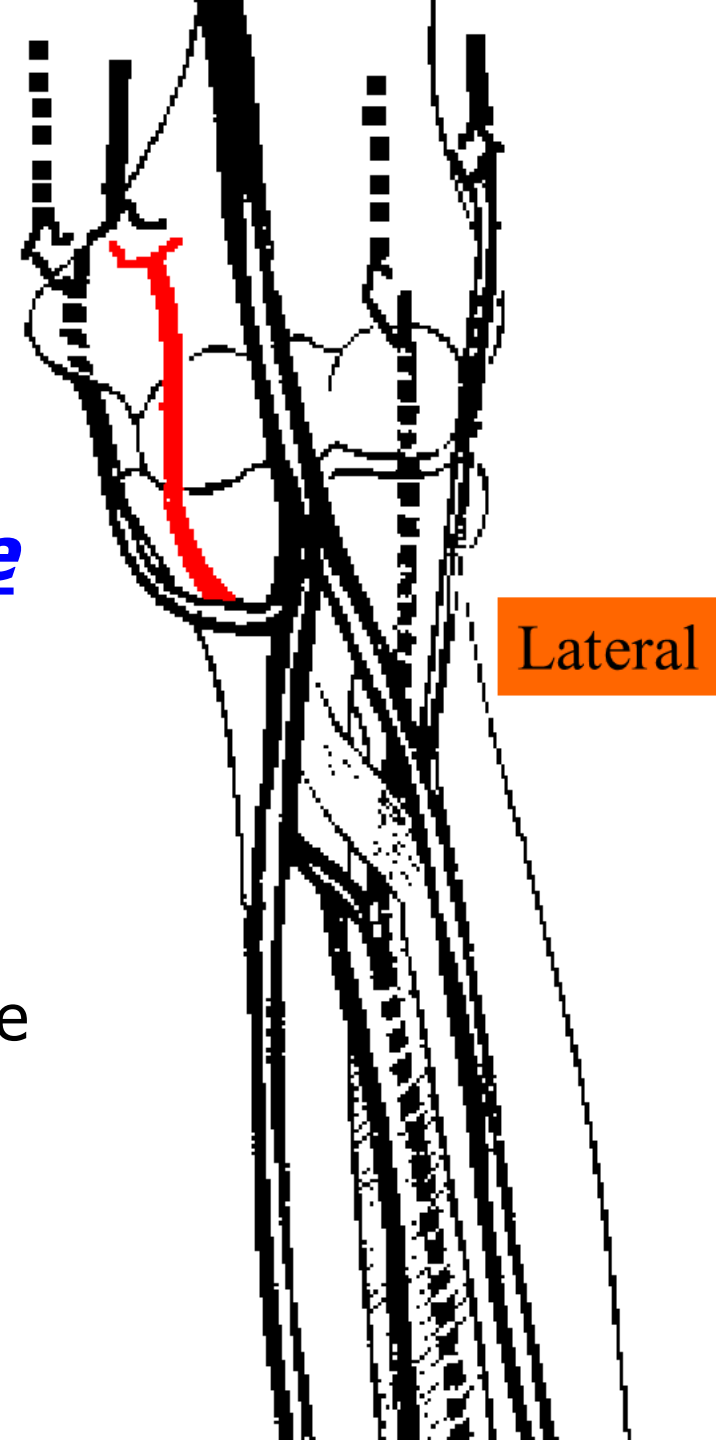
Formation

Formed by 8 branches

A) 4 around medial epicondyle of the humerus

Anteriorly

- 1) **anterior ulnar recurrent a.** from ulnar a.
 - ascends anterior to medial epicondyle
 - to anastomoses with
- 2) **inferior ulnar collateral a.** branch from brachial a.



Anastomosis around the elbow

A) 4 around medial epicondyle of the humerus

Posteriorly

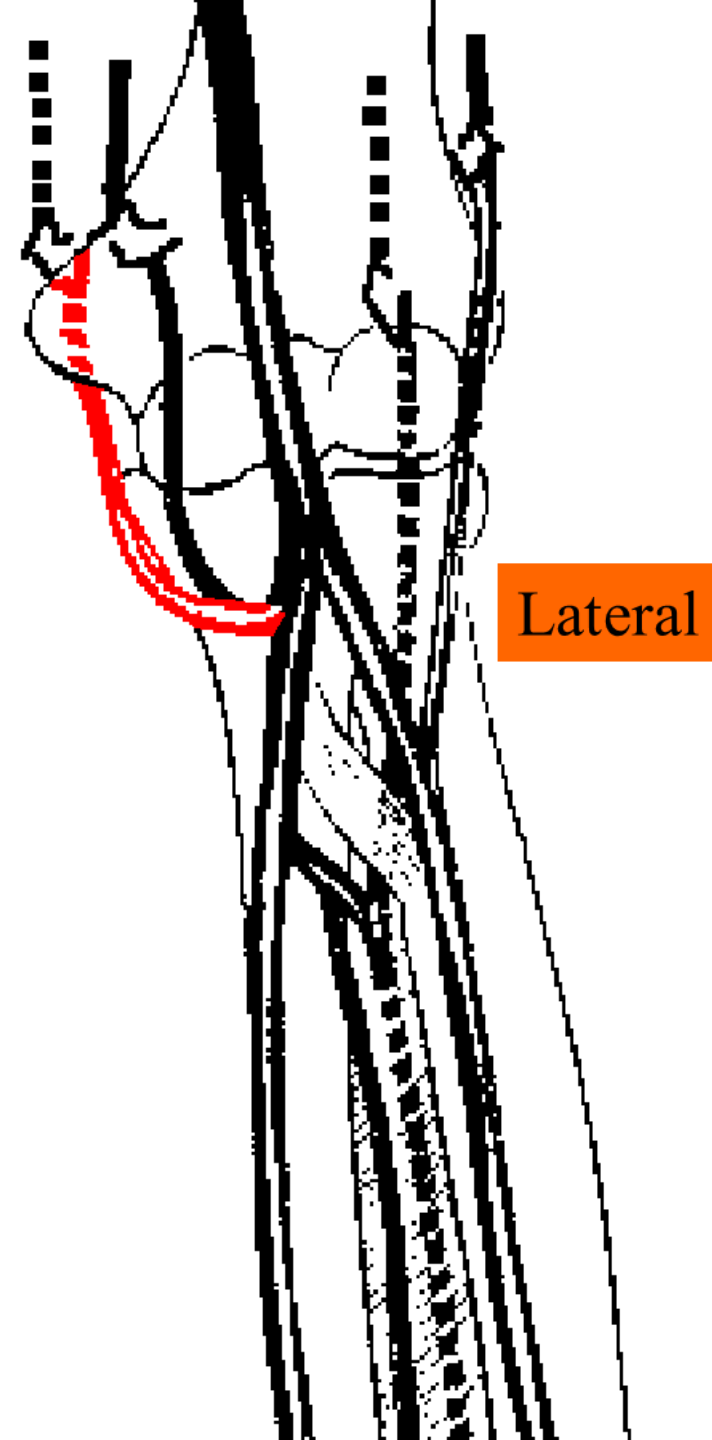
3) **posterior ulnar recurrent a.**

branch of ulnar a.

- ascends posterior to medial epicondyle
- to anastomose with

4) **superior ulnar collateral a.**

branch of brachial a.



Anastomosis around the elbow

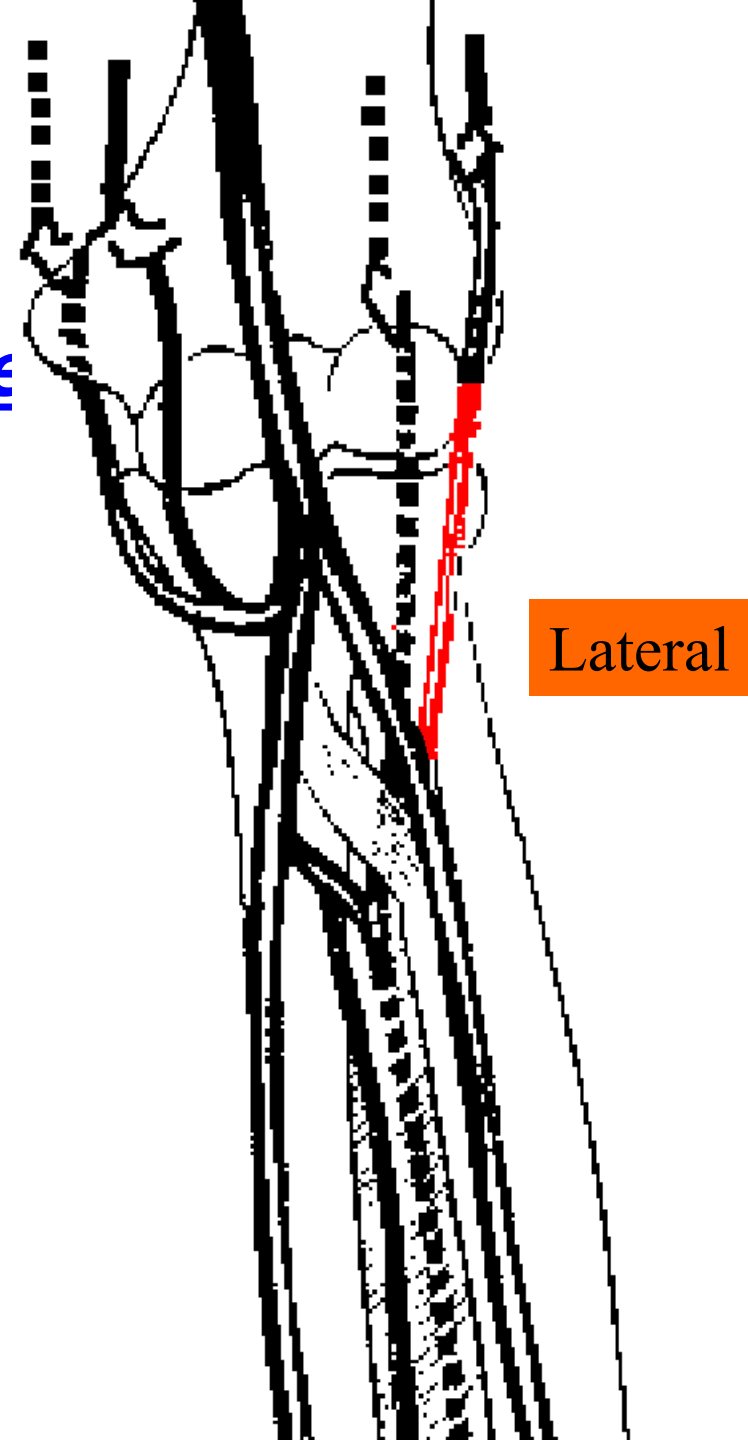
B) 4 around lateral epicondyle of the humerus

Anteriorly

1) **radial recurrent a.** from radial artery

- ascends anterior to lateral epicondyle
- to anastomoses with

2) **anterior descending branch** of the profunda brachii artery



Anastomosis around the elbow

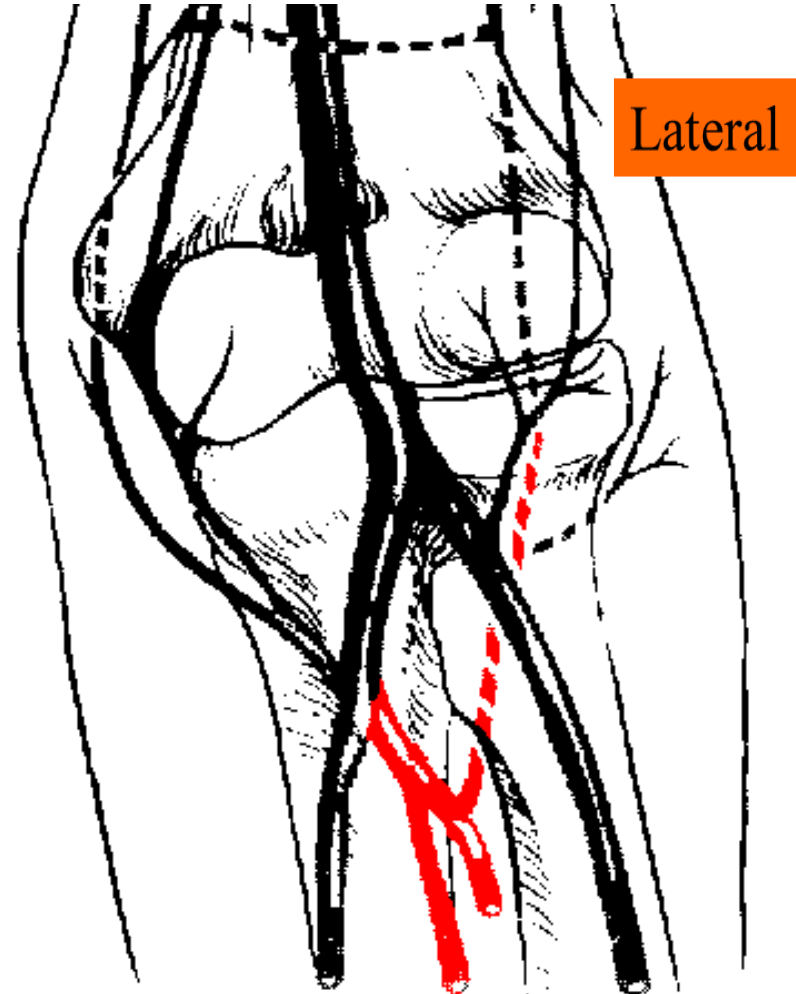
4 around lateral epicondyle of the humerus

Posteriorly

3) **posterior interosseous recurrent a.**

- Branch from posterior interosseous artery
- ascends posterior to the lateral epicondyle of the humerus
- to anastomose with

4) **posterior descending branch** of the profunda brachii a..



Anastomosis around elbow

